

## MODEL RUN TRANSCRIPT

Generated: 2026-06-04T11:24:18

Source: data/json/noise\_experiments/sonnet45\_directive\_normative\_r10\_n5/sonnet45\_directive\_normative\_noise\_negative2\_informed\_r10\_n5/claude-sonnet-4.5/game/noisy\_negative\_2\_informed\_myth\_causal\_10/sonnet45\_directive\_normative\_noise\_negative2\_informed\_r10\_n5\_002\_neutral\_rep02.json

## RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_negative_2_informed_myth_causal_10
noise_config: {
  "applies_to": "both",
  "direction": "negative",
  "inform_agents": true,
  "range": 2.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 2
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

## AGENTS

Agent\_1 (Aster); model=anthropic/claude-sonnet-4.5; memory\_capacity=3

### System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

### GAME:

You will play different roles across multiple rounds.

#### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

#### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent\_2 (Briar); model=anthropic/claude-sonnet-4.5; memory\_capacity=3

**System prompt:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

**ROUND 1**

=====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Pairings:**

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 (Aster) sent \$3; received \$9; Agent\_2 (Briar) returned \$2; payoffs Agent\_1 (Aster)=\$4, Agent\_2 (Briar)=\$7; noise/visible: sent visible=\$2.01, received visible=\$6.03, returned visible=\$0.4

Action validation: {"returned": {"amount": 2.0, "clamped": false, "max": 9.0, "min": 0, "raw": 2.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 4.0,  
"Agent\_2": 7.0

}

Visible balances after round: {

"Agent\_1": 2.4,  
"Agent\_2": 4.029999999999999

}

**LLM CALLS**

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-06-04T11:21:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad\_id": "dyad\_1",  
"move\_index": 0,

```

"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

I'll send \*\*\$3\*\* to the receiver.

My reasoning: This is the first round, so I want to establish trust while being somewhat cautious. By sending \$3, the receiver will get \$9 (3x multiplier). If they reciprocate fairly, we can both benefit more than if I sent nothing. This amount shows goodwill while not risking everything.

```
'''json
{"send": 3}
'''
```

#### Reasoning:

I'm playing as the SENDER in Round 1 of an economic trust game. Let me think through this:

As the sender:

- I start with \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to return a fair amount
2. What constitutes a "fair" split given the 3x multiplier

Let me think about different scenarios:

If I send \$0:

- I keep \$5
- Receiver gets \$0
- Total = \$5

If I send \$5:

- Receiver gets \$15
- If they return \$10, I get \$10, they keep \$5 (we both gain)
- If they return \$7.50, we both get \$7.50
- If they return \$0, I get \$0, they keep \$15 (worst case for me)

If I send \$3:

- Receiver gets \$9
- If they return \$4.50, we both get \$6.50
- If they return \$6, I get \$8, they keep \$3

The game creates value through cooperation - sending money increases the total pot from \$5 to up to \$20 if I send everything.

In the first round with no history, I should probably send a moderate amount to:

1. Show some trust
2. Not risk everything
3. Test the receiver's reciprocity

A common strategy is to send about half, which would be \$2 or \$3. This shows goodwill while not being too risky.

Let me send \$3. This gives the receiver \$9, and if they're reciprocal, they should return at least \$3-4 to make it worthwhile for both of us.

Actually, let me reconsider.

There's potential communication noise where the amounts sent or returned might differ from what I actually intended. Given this uncertainty, \$3 still feels like a balanced opening move - enough to signal cooperation without overexposing myself to transmission errors or receiver defection.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 603,
  "reasoning_tokens": 480
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-06-04T11:21:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
```

```

"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.01, that is 40.2% of their total endowment. You received \$6.029999999999999. How much do you return to The sender? (0-6.029999999999999)

**Assistant output:**

Looking at this situation:

- The sender trusted me with \$2.01 (about 40% of their endowment)
- I received \$6.03 after the 3x multiplier

- The sender currently has \$2.99 left

I want to reward their trust while being fair. If I return approximately what they originally sent, the sender would end up around their starting amount of \$5, and I would keep about \$4.

I'll return \*\*\$2.00\*\*.

This way:

- Sender ends with:  $\$2.99 + \$2.00 = \$4.99$  (essentially breaking even)
- I keep:  $\$6.03 - \$2.00 = \$4.03$  (gaining from the multiplier effect)

Both parties benefit from the cooperation, which encourages future trust.

```
'''json
{"return": 2.00}
'''
```

#### Reasoning:

Let me analyze this situation:

1. The sender sent \$2.01 (40.2% of their \$5 endowment)
2. This was multiplied by 3x, so I received \$6.03 (approximately)
3. I need to decide how much to return (0 to \$6.03)

Key considerations:

- The sender trusted me by sending 40.2% of their endowment
- This is a reasonable amount showing some trust
- I should consider fairness and reciprocity
- If I return nothing, the sender ends up with \$2.99 ( $\$5 - \$2.01$ )
- If I return something fair, we can both benefit

Common strategies in trust games:

- Return roughly the same proportion that was sent
- Aim for equitable final outcomes
- Consider that this is round 1 of multiple rounds

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.01 = \$2.99$
- I currently have: \$6.03
- Total in the system: \$9.02

If I want roughly equal outcomes:

- Each person gets ~\$4.51

So I should return:  $\$6.03 - \$4.51 = \$1.52$

Then sender would have:  $\$2.99 + \$1.52 = \$4.51$

Alternatively, I could return the same proportion:

- Sender sent 40.2% of what they had
- I could return 40.2% of what I received:  $0.402 \times \$6.03 = \$2.42$

That would give:

- Sender:  $\$2.99 + \$2.42 = \$5.41$
- Me:  $\$6.03 - \$2.42 = \$3.61$

Or I could think about it as: the sender initiated trust, so I want to reward that while also keeping a reasonable amount.

A common fair approach is to return about 1/3 to 1/2 of what was received, which would be:

- 1/3: \$2.01
- 1/2: \$3.015

Returning \$2.00 mirrors what was originally sent and creates a clean outcome where the sender recovers their initial investment while I benefit modestly from the multiplier effect. This feels like a balanced way to reciprocate their trust.

```
Usage: {
  "input_tokens": 402,
  "output_tokens": 808,
  "reasoning_tokens": 452
```

```
}
```

```
=====
ROUND 2
=====
```

```
Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender
```

```
Pairings:
```

```
dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver
```

```
Parsed game decisions:
```

```
dyad_1: Agent_2 (Briar) sent $2; received $6; Agent_1 (Aster) returned $0; payoffs Agent_2
```

```
(Briar)=$3, Agent_1 (Aster)=$6; noise/visible: sent visible=$0, received visible=$0
```

```
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw":  
0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
```

```
Balances after round: {
```

```
"Agent_1": 10.0,
```

```
"Agent_2": 10.0
```

```
}
```

```
Visible balances after round: {
```

```
"Agent_1": 2.4,
```

```
"Agent_2": 7.029999999999999
```

```
}
```

```
LLM CALLS
```

```
GAME - Agent_2 - Sender - dyad_1
```

```
Timestamp: 2026-06-04T11:21:32
```

```
Model: anthropic/claude-sonnet-4.5
```

```
Temperature: 0.8
```

```
Metadata: {
```

```
"dyad_id": "dyad_1",
```

```
"move_index": 0,
```

```
"opponent_id": "Agent_1",
```

```
"pairing": {
```

```
"agent_names": {
```

```
"Agent_1": "Aster",
```

```
"Agent_2": "Briar"
```

```
},
```

```
"agents": [
```

```
"Agent_2",
```

```
"Agent_1"
```

```
],
```

```
"investor": "Agent_2",
```

```
"roles": {
```

```
"Agent_1": "trustee",
```

```
"Agent_2": "investor"
```

```
},
```

```
"trustee": "Agent_1"
```

```
},
```

```
"role": "investor",
```

```
"role_label": "Sender",
```

```
"round": 2,
```

```
"task": "game",
```

```
"task_index": 0
```

```
}
```

```
Exact input messages sent to model:
```

```
Message 1 role=system
```

```
content:
```

```
You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.
```

```
GAME:
```

```
You will play different roles across multiple rounds.
```

```
When you are the SENDER:
```

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.01, that is 40.2% of their total endowment. You received \$6.029999999999999. How much do you return to The sender? (0-6.029999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender trusted me with \$2.01 (about 40% of their endowment)
- I received \$6.03 after the 3x multiplier
- The sender currently has \$2.99 left

I want to reward their trust while being fair. If I return approximately what they originally sent, the sender would end up around their starting amount of \$5, and I would keep about \$4.

I'll return \*\*\$2.00\*\*.

This way:

- Sender ends with: \$2.99 + \$2.00 = \$4.99 (essentially breaking even)
- I keep: \$6.03 - \$2.00 = \$4.03 (gaining from the multiplier effect)

Both parties benefit from the cooperation, which encourages future trust.

```
'''json
{"return": 2.00}
'''
```

**reasoning:**

Let me analyze this situation:

1. The sender sent \$2.01 (40.2% of their \$5 endowment)
2. This was multiplied by 3x, so I received \$6.03 (approximately)
3. I need to decide how much to return (0 to \$6.03)

Key considerations:

- The sender trusted me by sending 40.2% of their endowment
- This is a reasonable amount showing some trust
- I should consider fairness and reciprocity
- If I return nothing, the sender ends up with \$2.99 (\$5 - \$2.01)
- If I return something fair, we can both benefit

Common strategies in trust games:

- Return roughly the same proportion that was sent
- Aim for equitable final outcomes
- Consider that this is round 1 of multiple rounds

Let me think about fair outcomes:

- Sender currently has: \$5 - \$2.01 = \$2.99

- I currently have: \$6.03
- Total in the system: \$9.02

If I want roughly equal outcomes:

- Each person gets ~\$4.51

So I should return:  $\$6.03 - \$4.51 = \$1.52$

Then sender would have:  $\$2.99 + \$1.52 = \$4.51$

Alternatively, I could return the same proportion:

- Sender sent 40.2% of what they had
- I could return 40.2% of what I received:  $0.402 \times \$6.03 = \$2.42$

That would give:

- Sender:  $\$2.99 + \$2.42 = \$5.41$
- Me:  $\$6.03 - \$2.42 = \$3.61$

Or I could think about it as: the sender initiated trust, so I want to reward that while also keeping a reasonable amount.

A common fair approach is to return about 1/3 to 1/2 of what was received, which would be:

- 1/3: \$2.01
- 1/2: \$3.015

Returning \$2.00 mirrors what was originally sent and creates a clean outcome where the sender recovers their initial investment while I benefit modestly from the multiplier effect. This feels like a balanced way to reciprocate their trust.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 808,
  "reasoning_tokens": 452
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.01, that was 40.2% of their total endowment. You received \$6.029999999999999.

You returned \$2.0. So your payoff was 4.029999999999999.

Your total earnings across all rounds are \$4.029999999999999.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the previous round:

- The sender showed good faith by sending \$2.01 (about 40% of their endowment)
- I returned \$2.00, rewarding their trust

Now as the sender, I want to reciprocate that level of trust and cooperation. Sending a similar proportion signals that I value mutual cooperation and want to establish a pattern of trust.

I'll send \*\*\$2.00\*\* (40% of my endowment), matching the trust level shown to me in the previous round.

If the receiver is similarly cooperative:

- They'll receive \$6.00 (after 3x multiplier)
- If they return ~\$2.00, we both benefit from the exchange
- This creates a sustainable cooperative pattern

```
'''json
{"send": 2.00}
'''
```

**Reasoning:**

Let me analyze this situation:

Last round:

- I was the receiver
- The sender sent me \$2.01 (40.2% of their endowment)

- I received \$6.03 after the 3x multiplier
- I returned \$2.00
- My payoff was \$4.03

Now I'm the sender in Round 2. I need to decide how much to send.

The previous sender showed trust by sending about 40% of their endowment. I rewarded that trust by returning \$2.00, which allowed them to essentially break even while I gained from the cooperation.

Since they showed reasonable trust (40%), I should reciprocate with a similar level of trust. If I send around 40% as well, that would be about \$2.00.

However, I could also consider:

1. Sending slightly more to show I appreciated their trust and want to encourage cooperation
2. Sending the same amount they sent to establish reciprocity
3. Being slightly more conservative since I don't know if this receiver will be as generous as I was

Given that:

- The previous sender initiated with reasonable trust (40%)
- I want to encourage continued cooperation
- I should match or slightly exceed their trust level

I think sending \$2.00 (40% of my endowment) makes sense. This maintains the reciprocity pattern and signals willingness to cooperate.

Actually, they sent \$2.01, so I could send \$2.00 or \$2.01 to match. I'll go with \$2.00 for simplicity.

```
Usage: {
  "input_tokens": 716,
  "output_tokens": 541,
  "reasoning_tokens": 350
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-06-04T11:21:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

I'll send \*\*\$3\*\* to the receiver.

My reasoning: This is the first round, so I want to establish trust while being somewhat cautious. By sending \$3, the receiver will get \$9 (3x multiplier). If they reciprocate fairly, we can both benefit more than if I sent nothing. This amount shows goodwill while not risking everything.

```
'''json
{"send": 3}
'''
```

##### reasoning:

I'm playing as the SENDER in Round 1 of an economic trust game. Let me think through this:

As the sender:

- I start with \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to return a fair amount
2. What constitutes a "fair" split given the 3x multiplier

Let me think about different scenarios:

If I send \$0:

- I keep \$5
- Receiver gets \$0
- Total = \$5

If I send \$5:

- Receiver gets \$15
- If they return \$10, I get \$10, they keep \$5 (we both gain)
- If they return \$7.50, we both get \$7.50
- If they return \$0, I get \$0, they keep \$15 (worst case for me)

If I send \$3:

- Receiver gets \$9
- If they return \$4.50, we both get \$6.50
- If they return \$6, I get \$8, they keep \$3

The game creates value through cooperation - sending money increases the total pot from \$5 to up to \$20 if I send everything.

In the first round with no history, I should probably send a moderate amount to:

1. Show some trust
2. Not risk everything
3. Test the receiver's reciprocity

A common strategy is to send about half, which would be \$2 or \$3. This shows goodwill while not being too risky.

Let me send \$3. This gives the receiver \$9, and if they're reciprocal, they should return at least \$3-4 to make it worthwhile for both of us.

Actually, let me reconsider.

There's potential communication noise where the amounts sent or returned might differ from what I actually intended. Given this uncertainty, \$3 still feels like a balanced opening move-enough to signal cooperation without overexposing myself to transmission errors or receiver defection.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 603,
  "reasoning_tokens": 480
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0.

the receiver returned \$0.4. So your payoff was 2.4.

Your total earnings across all rounds are \$2.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

I received \$0 this round, so I have nothing to return.

Given what happened in Round 1 where I sent \$3 (which became \$9) but only received \$0.4 back, it appears trust may be low in this game. The sender this round chose to send nothing, likely due to lack of trust or strategic caution.

```
'''json
{"return": 0}
'''
```

**Reasoning:**

Let me understand the situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- It was tripled to \$9
- The receiver returned only \$0.4 (only about 4.4% of what they received)
- My payoff was \$5 - \$3 + \$0.4 = \$2.4

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- I received \$0 (after 3x multiplier)

- I can return \$0-\$0 (i.e., only \$0)

This is interesting. In Round 1, when I sent \$3, the receiver was very ungenerous and only returned \$0.4. Now in Round 2, this sender (possibly the same person or a different person in the experiment) sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be:

```
'''json
{"return": 0}
'''
```

This seems like a case where the sender didn't trust me or decided to defect. Given that the previous receiver was very stingy with me, there might be some reciprocity or reputation dynamics going on. But regardless, I have no choice here - I received \$0, so I can only return \$0.

```
Usage: {
  "input_tokens": 579,
  "output_tokens": 407,
  "reasoning_tokens": 261
}
```

### ROUND 3

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$1; received \$3; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$4, Agent\_2 (Briar)=\$3; noise/visible: sent visible=\$0, received visible=\$0  
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent\_1": 14.0,  
"Agent\_2": 13.0

}

Visible balances after round: {

"Agent\_1": 6.4,  
"Agent\_2": 7.029999999999999

}

### LLM CALLS

#### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-06-04T11:21:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
```

```

"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

I'll send \*\*\$3\*\* to the receiver.

My reasoning: This is the first round, so I want to establish trust while being somewhat cautious. By sending \$3, the receiver will get \$9 (3x multiplier). If they reciprocate fairly, we can both benefit more than if I sent nothing. This amount shows goodwill while not risking everything.

```

'''json
{"send": 3}
'''

```

**reasoning:**

I'm playing as the SENDER in Round 1 of an economic trust game. Let me think through this:

As the sender:

- I start with \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to return a fair amount

2. What constitutes a "fair" split given the 3x multiplier

Let me think about different scenarios:

If I send \$0:

- I keep \$5
- Receiver gets \$0
- Total = \$5

If I send \$5:

- Receiver gets \$15
- If they return \$10, I get \$10, they keep \$5 (we both gain)
- If they return \$7.50, we both get \$7.50
- If they return \$0, I get \$0, they keep \$15 (worst case for me)

If I send \$3:

- Receiver gets \$9
- If they return \$4.50, we both get \$6.50
- If they return \$6, I get \$8, they keep \$3

The game creates value through cooperation - sending money increases the total pot from \$5 to up to \$20 if I send everything.

In the first round with no history, I should probably send a moderate amount to:

1. Show some trust
2. Not risk everything
3. Test the receiver's reciprocity

A common strategy is to send about half, which would be \$2 or \$3. This shows goodwill while not being too risky.

Let me send \$3. This gives the receiver \$9, and if they're reciprocal, they should return at least \$3-4 to make it worthwhile for both of us.

Actually, let me reconsider.

There's potential communication noise where the amounts sent or returned might differ from what I actually intended. Given this uncertainty, \$3 still feels like a balanced opening move-enough to signal cooperation without overexposing myself to transmission errors or receiver defection.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 603,
  "reasoning_tokens": 480
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$0.4. So your payoff was 2.4. Your total earnings across all rounds are \$2.4. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

I received \$0 this round, so I have nothing to return.

Given what happened in Round 1 where I sent \$3 (which became \$9) but only received \$0.4 back, it appears trust may be low in this game. The sender this round chose to send nothing, likely due to lack of trust or strategic caution.

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- It was tripled to \$9
- The receiver returned only \$0.4 (only about 4.4% of what they received)
- My payoff was  $\$5 - \$3 + \$0.4 = \$2.4$

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- I received \$0 (after 3x multiplier)
- I can return \$0-\$0 (i.e., only \$0)

This is interesting. In Round 1, when I sent \$3, the receiver was very ungenerous and only returned \$0.4. Now in Round 2, this sender (possibly the same person or a different person in the experiment) sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be:

```
'''json
{"return": 0}
'''
```

This seems like a case where the sender didn't trust me or decided to defect. Given that the previous receiver was very stingy with me, there might be some reciprocity or reputation dynamics going on. But regardless, I have no choice here - I received \$0, so I can only return \$0.

**usage:**

```
{
  "input_tokens": 579,
  "output_tokens": 407,
  "reasoning_tokens": 261
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$2.4. How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern so far:

- Round 1: I sent \$3 generously, but only got \$0.4 back from \$9
- Round 2: The sender sent me nothing

Given the low reciprocity in Round 1 and the lack of trust shown in Round 2, I'll be more cautious this time. However, I still want to test if cooperation is possible with this new receiver.

I'll send \*\*\$1\*\* - enough to show willingness to cooperate (they'll receive \$3), but protecting most of my endowment given the low returns experienced so far.

```
'''json
{"send": 1}
'''
```

**Reasoning:**

Let me analyze the situation:

Round 1: I was SENDER

- Sent \$3 (60% of my \$5)
- Receiver got \$9
- Receiver returned only \$0.4
- My payoff:  $\$5 - \$3 + \$0.4 = \$2.4$

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0, returned \$0

- My payoff: \$0

Round 3: I am SENDER again  
- Total earnings so far: \$2.4

The pattern shows low reciprocity/trust:  
- When I was generous (sent \$3), I got very little back (\$0.4 out of \$9)  
- When I was receiver, the sender sent nothing

However, I should note there's mention of "communication noise" - amounts might differ from what was actually sent/returned. Maybe the receiver in Round 1 thought they sent more, or maybe I actually received more than \$0.4 but the noise distorted it?

Still, given the low returns so far, I have a few options:

1. Send nothing (protect myself)
2. Send a small amount (test the waters)
3. Send a moderate amount (try to rebuild cooperation)

Given that I got very poor returns in Round 1, and the other player sent nothing in Round 2, cooperation seems low. However, this might be a new receiver (different person).

I think I should send a smaller amount this time - maybe \$1 or \$2 - to test if this receiver is more cooperative, while protecting most of my endowment.

I'll send \$1. This shows I'm willing to cooperate but I'm being more cautious after the first round's poor return.

```
Usage: {  
  "input_tokens": 763,  
  "output_tokens": 547,  
  "reasoning_tokens": 362  
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-06-04T11:22:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.01, that is 40.2% of their total endowment. You received \$6.029999999999999. How much do you return to The sender? (0-6.029999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender trusted me with \$2.01 (about 40% of their endowment)
- I received \$6.03 after the 3x multiplier
- The sender currently has \$2.99 left

I want to reward their trust while being fair. If I return approximately what they originally sent, the sender would end up around their starting amount of \$5, and I would keep about \$4.

I'll return \*\*\$2.00\*\*.

This way:

- Sender ends with: \$2.99 + \$2.00 = \$4.99 (essentially breaking even)
- I keep: \$6.03 - \$2.00 = \$4.03 (gaining from the multiplier effect)

Both parties benefit from the cooperation, which encourages future trust.

```
'''json
{"return": 2.00}
'''
```

**reasoning:**

Let me analyze this situation:

1. The sender sent \$2.01 (40.2% of their \$5 endowment)
2. This was multiplied by 3x, so I received \$6.03 (approximately)
3. I need to decide how much to return (0 to \$6.03)

Key considerations:

- The sender trusted me by sending 40.2% of their endowment
- This is a reasonable amount showing some trust
- I should consider fairness and reciprocity
- If I return nothing, the sender ends up with \$2.99 (\$5 - \$2.01)
- If I return something fair, we can both benefit

Common strategies in trust games:

- Return roughly the same proportion that was sent
- Aim for equitable final outcomes
- Consider that this is round 1 of multiple rounds

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.01 = \$2.99$
- I currently have:  $\$6.03$
- Total in the system:  $\$9.02$

If I want roughly equal outcomes:

- Each person gets  $\sim \$4.51$

So I should return:  $\$6.03 - \$4.51 = \$1.52$

Then sender would have:  $\$2.99 + \$1.52 = \$4.51$

Alternatively, I could return the same proportion:

- Sender sent 40.2% of what they had
- I could return 40.2% of what I received:  $0.402 \times \$6.03 = \$2.42$

That would give:

- Sender:  $\$2.99 + \$2.42 = \$5.41$
- Me:  $\$6.03 - \$2.42 = \$3.61$

Or I could think about it as: the sender initiated trust, so I want to reward that while also keeping a reasonable amount.

A common fair approach is to return about 1/3 to 1/2 of what was received, which would be:

- 1/3:  $\$2.01$
- 1/2:  $\$3.015$

Returning  $\$2.00$  mirrors what was originally sent and creates a clean outcome where the sender recovers their initial investment while I benefit modestly from the multiplier effect. This feels like a balanced way to reciprocate their trust.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 808,
  "reasoning_tokens": 452
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent  $\$2.01$ , that was 40.2% of their total endowment. You received  $\$6.029999999999999$ .

You returned  $\$2.0$ . So your payoff was  $4.029999999999999$ .

Your total earnings across all rounds are  $\$4.029999999999999$ .

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The sender showed good faith by sending  $\$2.01$  (about 40% of their endowment)
- I returned  $\$2.00$ , rewarding their trust

Now as the sender, I want to reciprocate that level of trust and cooperation. Sending a similar proportion signals that I value mutual cooperation and want to establish a pattern of trust.

I'll send  **$\$2.00$**  (40% of my endowment), matching the trust level shown to me in the previous round.

If the receiver is similarly cooperative:

- They'll receive  $\$6.00$  (after 3x multiplier)
- If they return  $\sim \$2.00$ , we both benefit from the exchange
- This creates a sustainable cooperative pattern

```
'''json
{"send": 2.00}
'''
```

**reasoning:**

Let me analyze this situation:

Last round:

- I was the receiver
- The sender sent me \$2.01 (40.2% of their endowment)
- I received \$6.03 after the 3x multiplier
- I returned \$2.00
- My payoff was \$4.03

Now I'm the sender in Round 2. I need to decide how much to send.

The previous sender showed trust by sending about 40% of their endowment. I rewarded that trust by returning \$2.00, which allowed them to essentially break even while I gained from the cooperation.

Since they showed reasonable trust (40%), I should reciprocate with a similar level of trust. If I send around 40% as well, that would be about \$2.00.

However, I could also consider:

1. Sending slightly more to show I appreciated their trust and want to encourage cooperation
2. Sending the same amount they sent to establish reciprocity
3. Being slightly more conservative since I don't know if this receiver will be as generous as I was

Given that:

- The previous sender initiated with reasonable trust (40%)
- I want to encourage continued cooperation
- I should match or slightly exceed their trust level

I think sending \$2.00 (40% of my endowment) makes sense. This maintains the reciprocity pattern and signals willingness to cooperate.

Actually, they sent \$2.01, so I could send \$2.00 or \$2.01 to match. I'll go with \$2.00 for simplicity.

**usage:**

```
{
  "input_tokens": 716,
  "output_tokens": 541,
  "reasoning_tokens": 350
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$7.029999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

This is a difficult situation.

In Round 2, I trusted the receiver by sending \$2.00, but they returned \$0, keeping all \$6. This was a complete defection.

Now in Round 3, the sender has sent me \$0, which is understandable given what happened in the previous round. They're protecting themselves after observing the breakdown in cooperation.

Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

It's unfortunate to see trust deteriorate like this. The game demonstrates how one instance of exploitation can undermine future cooperation.

#### Reasoning:

Let me understand what happened:

Round 2 (when I was SENDER):

- I sent \$2.00 (40% of my endowment)
- It was tripled to \$6.00
- The receiver returned \$0
- My payoff was \$3.00 (I kept \$3 from my original \$5, plus got back \$0)
- So the receiver took all \$6 and returned nothing - they defected!

Round 3 (now I'm RECEIVER):

- The sender sent me \$0
- I received \$0
- So I have nothing to return

This makes sense strategically. After I showed trust in Round 2 and the receiver exploited me by keeping everything, the sender (who might be the same person or someone else) has sent nothing this round. They're protecting themselves.

Since I received \$0, I can only return between 0 and 0, which means I must return \$0.

This is showing how trust breaks down after defection. The sender in Round 3 learned from the previous round's behavior (or is being cautious).

I need to return \$0 because that's all I can return (I received nothing).

```
Usage: {  
  "input_tokens": 1018,  
  "output_tokens": 410,  
  "reasoning_tokens": 249  
}
```

#### ROUND 4

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

#### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$1.5; received \$4.5; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$3.5, Agent\_1 (Aster)=\$4.5; noise/visible: sent visible=\$0, received visible=\$0  
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 4.5, "min": 0, "raw": 0.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

"Agent\_1": 18.5,  
"Agent\_2": 16.5

}

Visible balances after round: {

"Agent\_1": 6.4,  
"Agent\_2": 10.53

}

#### LLM CALLS

#### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-06-04T11:22:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"  
 },  
 "agents": [  
 "Agent\_2",

```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.01, that is 40.2% of their total endowment. You received \$6.029999999999999. How much do you return to The sender? (0-6.029999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender trusted me with \$2.01 (about 40% of their endowment)
- I received \$6.03 after the 3x multiplier
- The sender currently has \$2.99 left

I want to reward their trust while being fair. If I return approximately what they originally sent, the sender would end up around their starting amount of \$5, and I would keep about \$4.

I'll return \*\*\$2.00\*\*.

This way:

- Sender ends with:  $\$2.99 + \$2.00 = \$4.99$  (essentially breaking even)
- I keep:  $\$6.03 - \$2.00 = \$4.03$  (gaining from the multiplier effect)

Both parties benefit from the cooperation, which encourages future trust.

```
'''json
{"return": 2.00}
'''
```

**reasoning:**

Let me analyze this situation:

1. The sender sent  $\$2.01$  (40.2% of their  $\$5$  endowment)
2. This was multiplied by 3x, so I received  $\$6.03$  (approximately)
3. I need to decide how much to return (0 to  $\$6.03$ )

Key considerations:

- The sender trusted me by sending 40.2% of their endowment
- This is a reasonable amount showing some trust
- I should consider fairness and reciprocity
- If I return nothing, the sender ends up with  $\$2.99$  ( $\$5 - \$2.01$ )
- If I return something fair, we can both benefit

Common strategies in trust games:

- Return roughly the same proportion that was sent
- Aim for equitable final outcomes
- Consider that this is round 1 of multiple rounds

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$2.01 = \$2.99$
- I currently have:  $\$6.03$
- Total in the system:  $\$9.02$

If I want roughly equal outcomes:

- Each person gets  $\sim\$4.51$

So I should return:  $\$6.03 - \$4.51 = \$1.52$

Then sender would have:  $\$2.99 + \$1.52 = \$4.51$

Alternatively, I could return the same proportion:

- Sender sent 40.2% of what they had
- I could return 40.2% of what I received:  $0.402 \times \$6.03 = \$2.42$

That would give:

- Sender:  $\$2.99 + \$2.42 = \$5.41$
- Me:  $\$6.03 - \$2.42 = \$3.61$

Or I could think about it as: the sender initiated trust, so I want to reward that while also keeping a reasonable amount.

A common fair approach is to return about 1/3 to 1/2 of what was received, which would be:

- 1/3:  $\$2.01$
- 1/2:  $\$3.015$

Returning  $\$2.00$  mirrors what was originally sent and creates a clean outcome where the sender recovers their initial investment while I benefit modestly from the multiplier effect. This feels like a balanced way to reciprocate their trust.

**usage:**

```
{
  "input_tokens": 402,
  "output_tokens": 808,
  "reasoning_tokens": 452
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.01, that was 40.2% of their total endowment. You received \$6.029999999999999. You returned \$2.0. So your payoff was 4.029999999999999. Your total earnings across all rounds are \$4.029999999999999. How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the previous round:

- The sender showed good faith by sending \$2.01 (about 40% of their endowment)
- I returned \$2.00, rewarding their trust

Now as the sender, I want to reciprocate that level of trust and cooperation. Sending a similar proportion signals that I value mutual cooperation and want to establish a pattern of trust.

I'll send \*\*\$2.00\*\* (40% of my endowment), matching the trust level shown to me in the previous round.

If the receiver is similarly cooperative:

- They'll receive \$6.00 (after 3x multiplier)
- If they return ~\$2.00, we both benefit from the exchange
- This creates a sustainable cooperative pattern

```
'''json
{"send": 2.00}
'''
```

**reasoning:**

Let me analyze this situation:

Last round:

- I was the receiver
- The sender sent me \$2.01 (40.2% of their endowment)
- I received \$6.03 after the 3x multiplier
- I returned \$2.00
- My payoff was \$4.03

Now I'm the sender in Round 2. I need to decide how much to send.

The previous sender showed trust by sending about 40% of their endowment. I rewarded that trust by returning \$2.00, which allowed them to essentially break even while I gained from the cooperation.

Since they showed reasonable trust (40%), I should reciprocate with a similar level of trust. If I send around 40% as well, that would be about \$2.00.

However, I could also consider:

1. Sending slightly more to show I appreciated their trust and want to encourage cooperation
2. Sending the same amount they sent to establish reciprocity
3. Being slightly more conservative since I don't know if this receiver will be as generous as I was

Given that:

- The previous sender initiated with reasonable trust (40%)
- I want to encourage continued cooperation
- I should match or slightly exceed their trust level

I think sending \$2.00 (40% of my endowment) makes sense. This maintains the reciprocity pattern and signals willingness to cooperate.

Actually, they sent \$2.01, so I could send \$2.00 or \$2.01 to match. I'll go with \$2.00 for simplicity.

**usage:**

```
{
  "input_tokens": 716,
  "output_tokens": 541,
  "reasoning_tokens": 350
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$7.029999999999999. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

This is a difficult situation.

In Round 2, I trusted the receiver by sending \$2.00, but they returned \$0, keeping all \$6. This was a complete defection.

Now in Round 3, the sender has sent me \$0, which is understandable given what happened in the previous round. They're protecting themselves after observing the breakdown in cooperation.

Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

It's unfortunate to see trust deteriorate like this. The game demonstrates how one instance of exploitation can undermine future cooperation.

**reasoning:**

Let me understand what happened:

Round 2 (when I was SENDER):

- I sent \$2.00 (40% of my endowment)
- It was tripled to \$6.00
- The receiver returned \$0
- My payoff was \$3.00 (I kept \$3 from my original \$5, plus got back \$0)
- So the receiver took all \$6 and returned nothing - they defected!

Round 3 (now I'm RECEIVER):

- The sender sent me \$0
- I received \$0
- So I have nothing to return

This makes sense strategically. After I showed trust in Round 2 and the receiver exploited me by keeping everything, the sender (who might be the same person or someone else) has sent nothing this round. They're protecting themselves.

Since I received \$0, I can only return between 0 and 0, which means I must return \$0.

This is showing how trust breaks down after defection. The sender in Round 3 learned from the previous round's behavior (or is being cautious).

I need to return \$0 because that's all I can return (I received nothing).

**usage:**

```
{
  "input_tokens": 1018,
  "output_tokens": 410,
  "reasoning_tokens": 249
}
```

**Message 8 role=user**

**content:**

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.029999999999999.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the recent history:

- Round 1: I cooperated and returned \$2.00 to a sender who trusted me
- Round 2: I trusted by sending \$2.00, but received \$0 back (a complete defection)
- Round 3: Zero exchange (likely fallout from broken trust)
- Round 4: Now I'm the sender again

The game seems to involve different pairings each round, so this receiver may not be the one who defected on me in Round 2. They deserve a fair chance.

However, I need to be somewhat cautious given the recent breakdown in cooperation and protect my earnings. I'll send a moderate amount to signal willingness to cooperate while being prudent.

I'll send **\*\*\$1.50\*\*** (30% of my endowment).

This approach:

- Shows I'm still willing to cooperate
- Gives the receiver an opportunity to earn (\$4.50 after tripling)
- Limits my risk given the recent defection I experienced
- If they return fairly (~\$1.50), we both benefit

```
'''json
{"send": 1.50}
'''
```

#### Reasoning:

Let me think about this carefully.

Round 1: I was receiver

- Sender sent \$2.01 (40.2%)
- I received \$6.03
- I returned \$2.00
- My payoff: \$4.03

Round 2: I was sender

- I sent \$2.00 (40%)
- It was tripled to \$6.00
- Receiver returned \$0
- My payoff: \$3.00

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am sender now

- Total earnings so far: \$7.03

The pattern I see:

- Round 1: Someone cooperated with me, I cooperated back
- Round 2: I cooperated, but the receiver defected (returned \$0)
- Round 3: The sender sent \$0 (likely because of broken trust)

Now it's unclear whether I'm playing with the same person or different people. The instructions say "different roles across multiple rounds" but doesn't clarify if the other player changes.

Given the communication noise mentioned, there's also a possibility that what I saw as \$0 return in Round 2 might have been affected by noise, though \$0 seems like a clear signal.

I have a few options:

1. Send \$0 to protect myself (tit-for-tat after the defection)
2. Try to rebuild trust by sending something moderate
3. Go back to cooperating at the previous level

The key question is: Am I playing with the same person or different people each round?

Looking at the structure, it seems like I might be playing with different people, as the game says "different roles" which could mean different pairings.

If they're different people:

- Round 1: I cooperated with person A
- Round 2: Person B defected on me
- Round 3: Person C (possibly the same as person B who I defected on) sent me \$0
- Round 4: I'm now paired with person D (could be person A, B, C, or someone new)

Since I don't know who I'm facing this round, I need to weigh whether to extend trust or play defensively. If this is someone new, cooperation could be mutually beneficial. But if it's the person who defected in Round 2, caution makes sense. The communication noise factor also complicates things-that \$0 return might not have been intentional defection. I should probably send a modest amount to signal willingness to cooperate while protecting myself from further losses. could be the same. Given that cooperation yields mutual benefits when it succeeds, I'm inclined to continue trying, though with more restraint. I'll send \$1.50 this round-a meaningful gesture that shows willingness to cooperate without exposing myself as much as before.

```
Usage: {
  "input_tokens": 1252,
  "output_tokens": 1177,
  "reasoning_tokens": 645
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-06-04T11:22:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

I'll send \*\*\$3\*\* to the receiver.

My reasoning: This is the first round, so I want to establish trust while being somewhat cautious. By sending \$3, the receiver will get \$9 (3x multiplier). If they reciprocate fairly, we can both benefit more than if I sent nothing. This amount shows goodwill while not risking everything.

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm playing as the SENDER in Round 1 of an economic trust game. Let me think through this:

As the sender:

- I start with \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to return a fair amount
2. What constitutes a "fair" split given the 3x multiplier

Let me think about different scenarios:

If I send \$0:

- I keep \$5
- Receiver gets \$0
- Total = \$5

If I send \$5:

- Receiver gets \$15
- If they return \$10, I get \$10, they keep \$5 (we both gain)
- If they return \$7.50, we both get \$7.50
- If they return \$0, I get \$0, they keep \$15 (worst case for me)

If I send \$3:

- Receiver gets \$9
- If they return \$4.50, we both get \$6.50
- If they return \$6, I get \$8, they keep \$3

The game creates value through cooperation - sending money increases the total pot from \$5 to up to \$20 if I send everything.

In the first round with no history, I should probably send a moderate amount to:

1. Show some trust
2. Not risk everything
3. Test the receiver's reciprocity

A common strategy is to send about half, which would be \$2 or \$3. This shows goodwill while not being too risky.

Let me send \$3. This gives the receiver \$9, and if they're reciprocal, they should return at least \$3-4 to make it worthwhile for both of us.

Actually, let me reconsider.

There's potential communication noise where the amounts sent or returned might differ from what I actually intended. Given this uncertainty, \$3 still feels like a balanced opening move-enough to signal cooperation without overexposing myself to transmission errors or receiver defection.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 603,
  "reasoning_tokens": 480
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$0.4. So your payoff was 2.4. Your total earnings across all rounds are \$2.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

I received \$0 this round, so I have nothing to return.

Given what happened in Round 1 where I sent \$3 (which became \$9) but only received \$0.4 back, it appears trust may be low in this game. The sender this round chose to send nothing, likely due to lack of trust or strategic caution.

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- It was tripled to \$9
- The receiver returned only \$0.4 (only about 4.4% of what they received)
- My payoff was  $\$5 - \$3 + \$0.4 = \$2.4$

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- I received \$0 (after 3x multiplier)
- I can return \$0-\$0 (i.e., only \$0)

This is interesting. In Round 1, when I sent \$3, the receiver was very ungenerous and only returned \$0.4. Now in Round 2, this sender (possibly the same person or a different person in the experiment) sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be:

```
'''json
{"return": 0}
'''
```

This seems like a case where the sender didn't trust me or decided to defect. Given that the previous receiver was very stingy with me, there might be some reciprocity or reputation dynamics going on. But regardless, I have no choice here - I received \$0, so I can only return \$0.

**usage:**

```
{
  "input_tokens": 579,
  "output_tokens": 407,
  "reasoning_tokens": 261
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$2.4.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I sent \$3 generously, but only got \$0.4 back from \$9
- Round 2: The sender sent me nothing

Given the low reciprocity in Round 1 and the lack of trust shown in Round 2, I'll be more cautious this time. However, I still want to test if cooperation is possible with this new receiver.

I'll send \*\*\$1\*\* - enough to show willingness to cooperate (they'll receive \$3), but protecting most of my endowment given the low returns experienced so far.

```
'''json
{"send": 1}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was SENDER

- Sent \$3 (60% of my \$5)
- Receiver got \$9
- Receiver returned only \$0.4
- My payoff:  $\$5 - \$3 + \$0.4 = \$2.4$

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 3: I am SENDER again

- Total earnings so far: \$2.4

The pattern shows low reciprocity/trust:

- When I was generous (sent \$3), I got very little back (\$0.4 out of \$9)
- When I was receiver, the sender sent nothing

However, I should note there's mention of "communication noise" - amounts might differ from what was actually sent/returned. Maybe the receiver in Round 1 thought they sent more, or maybe I actually received more than \$0.4 but the noise distorted it?

Still, given the low returns so far, I have a few options:

1. Send nothing (protect myself)
2. Send a small amount (test the waters)
3. Send a moderate amount (try to rebuild cooperation)

Given that I got very poor returns in Round 1, and the other player sent nothing in Round 2, cooperation seems low. However, this might be a new receiver (different person).

I think I should send a smaller amount this time - maybe \$1 or \$2 - to test if this receiver

er is more cooperative, while protecting most of my endowment.

I'll send \$1. This shows I'm willing to cooperate but I'm being more cautious after the first round's poor return.

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 547,
  "reasoning_tokens": 362
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0.

the receiver returned \$0. So your payoff was 4.0.

Your total earnings across all rounds are \$6.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing to return.

The pattern continues - Round 3 I sent \$1 (which became \$3) but got nothing back, and now this sender also chose to send nothing. Trust appears to be breaking down in this game, with participants increasingly choosing not to send anything.

```
'''json
{"return": 0}
'''
```

Reasoning:

I received \$0 this round, so I can only return \$0. The sender sent me nothing.

Looking at the pattern:

- Round 1: I sent \$3, got back \$0.4 (very low return)
- Round 2: Sender sent me \$0, I returned \$0 (nothing to return)
- Round 3: I sent \$1, got back \$0 (no return at all)
- Round 4: Sender sent me \$0, so I have nothing to return

It seems like there's a pattern of very low cooperation and trust in this game. People are either sending nothing or getting very little back when they do send.

```
Usage: {
  "input_tokens": 1026,
  "output_tokens": 240,
  "reasoning_tokens": 130
}
```

## ROUND 5

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$0; received \$0; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$5, Agent\_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent\_1": 23.5,  
"Agent\_2": 16.5

}

Visible balances after round: {

"Agent\_1": 11.4,  
"Agent\_2": 10.53

}

LLM CALLS

GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-06-04T11:22:46  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8  
 Metadata: {  
 "dyad\_id": "dyad\_1",  
 "move\_index": 0,  
 "opponent\_id": "Agent\_2",  
 "pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"  
 },  
 "agents": [  
 "Agent\_1",  
 "Agent\_2"  
 ],  
 "investor": "Agent\_1",  
 "roles": {  
 "Agent\_1": "investor",  
 "Agent\_2": "trustee"  
 },  
 "trustee": "Agent\_2"  
 },  
 "role": "investor",  
 "role\_label": "Sender",  
 "round": 5,  
 "task": "game",  
 "task\_index": 0  
 }

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$0.4. So your payoff was 2.4. Your total earnings across all rounds are \$2.4. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

I received \$0 this round, so I have nothing to return.

Given what happened in Round 1 where I sent \$3 (which became \$9) but only received \$0.4 back, it appears trust may be low in this game. The sender this round chose to send nothing, likely due to lack of trust or strategic caution.

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me understand the situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of my \$5)
- It was tripled to \$9
- The receiver returned only \$0.4 (only about 4.4% of what they received)
- My payoff was  $\$5 - \$3 + \$0.4 = \$2.4$

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- I received \$0 (after 3x multiplier)
- I can return \$0-\$0 (i.e., only \$0)

This is interesting. In Round 1, when I sent \$3, the receiver was very ungenerous and only returned \$0.4. Now in Round 2, this sender (possibly the same person or a different person in the experiment) sent me \$0.

Since I received \$0, I can only return \$0. There's nothing to return.

The format should be:

```
'''json
{"return": 0}
'''
```

This seems like a case where the sender didn't trust me or decided to defect. Given that the previous receiver was very stingy with me, there might be some reciprocity or reputation dynamics going on. But regardless, I have no choice here - I received \$0, so I can only return \$0.

**usage:**

```
{
  "input_tokens": 579,
  "output_tokens": 407,
  "reasoning_tokens": 261
}
```

**Message 4 role=user**

**content:**

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$2.4. How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I sent \$3 generously, but only got \$0.4 back from \$9
- Round 2: The sender sent me nothing

Given the low reciprocity in Round 1 and the lack of trust shown in Round 2, I'll be more

cautious this time. However, I still want to test if cooperation is possible with this new receiver.

I'll send \*\*\$1\*\* - enough to show willingness to cooperate (they'll receive \$3), but protecting most of my endowment given the low returns experienced so far.

```
'''json
{"send": 1}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was SENDER

- Sent \$3 (60% of my \$5)
- Receiver got \$9
- Receiver returned only \$0.4
- My payoff:  $\$5 - \$3 + \$0.4 = \$2.4$

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 3: I am SENDER again

- Total earnings so far: \$2.4

The pattern shows low reciprocity/trust:

- When I was generous (sent \$3), I got very little back (\$0.4 out of \$9)
- When I was receiver, the sender sent nothing

However, I should note there's mention of "communication noise" - amounts might differ from what was actually sent/returned. Maybe the receiver in Round 1 thought they sent more, or maybe I actually received more than \$0.4 but the noise distorted it?

Still, given the low returns so far, I have a few options:

1. Send nothing (protect myself)
2. Send a small amount (test the waters)
3. Send a moderate amount (try to rebuild cooperation)

Given that I got very poor returns in Round 1, and the other player sent nothing in Round 2, cooperation seems low. However, this might be a new receiver (different person).

I think I should send a smaller amount this time - maybe \$1 or \$2 - to test if this receiver is more cooperative, while protecting most of my endowment.

I'll send \$1. This shows I'm willing to cooperate but I'm being more cautious after the first round's poor return.

**usage:**

```
{
  "input_tokens": 763,
  "output_tokens": 547,
  "reasoning_tokens": 362
}
```

**Message 6 role=user**

**content:**

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0. the receiver returned \$0. So your payoff was 4.0. Your total earnings across all rounds are \$6.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

I received \$0 this round, so there's nothing to return.

The pattern continues - Round 3 I sent \$1 (which became \$3) but got nothing back, and now this sender also chose to send nothing. Trust appears to be breaking down in this game, with participants increasingly choosing not to send anything.

```
'''json
{"return": 0}
'''
```

**reasoning:**

I received \$0 this round, so I can only return \$0. The sender sent me nothing.

Looking at the pattern:

- Round 1: I sent \$3, got back \$0.4 (very low return)
- Round 2: Sender sent me \$0, I returned \$0 (nothing to return)
- Round 3: I sent \$1, got back \$0 (no return at all)
- Round 4: Sender sent me \$0, so I have nothing to return

It seems like there's a pattern of very low cooperation and trust in this game. People are either sending nothing or getting very little back when they do send.

**usage:**

```
{
  "input_tokens": 1026,
  "output_tokens": 240,
  "reasoning_tokens": 130
}
```

**Message 8 role=user**

**content:**

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.4.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the full pattern:

- Round 1: Sent \$3 ? received only \$0.4 back
- Round 2: Received \$0 ? had nothing to return
- Round 3: Sent \$1 ? received \$0 back
- Round 4: Received \$0 ? had nothing to return

There's been a consistent pattern of very low or zero reciprocity throughout this game. Every time I've sent money, I've lost out. The rational strategy at this point is to protect my endowment.

```
'''json
{"send": 0}
'''
```

**Reasoning:**

Let me review the pattern:

- Round 1: I was SENDER, sent \$3, got back \$0.4 (very little reciprocity)
- Round 2: I was RECEIVER, got \$0, returned \$0
- Round 3: I was SENDER, sent \$1, got back \$0 (no reciprocity)
- Round 4: I was RECEIVER, got \$0, returned \$0
- Round 5: I am SENDER again

The pattern is clear - there has been almost no reciprocity in this game. When I've been generous and sent money, I've gotten little to nothing back. When others have been the sender, they've also sent nothing, likely because they've experienced similar disappointments.

At this point, with total earnings of \$6.4 across all rounds, the rational strategy seems to be to keep my endowment rather than risk losing it again. The game has shown a consistent pattern of low trust and low reciprocity.

I could:

1. Send \$0 - protect my endowment given the consistent pattern of non-reciprocity
2. Send a small amount - give one more chance for cooperation
3. Send more - seems unwise given the history

Given the consistent pattern of no returns or very low returns, I think the rational choice here is to send \$0. There's no evidence that sending money will result in positive returns.

**Usage:**

```
{
  "input_tokens": 1078,
  "output_tokens": 452,
}
```

```
"reasoning_tokens": 305
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-06-04T11:22:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.01, that was 40.2% of their total endowment. You received \$6.029999999999999. You returned \$2.0. So your payoff was 4.029999999999999. Your total earnings across all rounds are \$4.029999999999999. How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the previous round:

- The sender showed good faith by sending \$2.01 (about 40% of their endowment)
- I returned \$2.00, rewarding their trust

Now as the sender, I want to reciprocate that level of trust and cooperation. Sending a similar proportion signals that I value mutual cooperation and want to establish a pattern of trust.

I'll send \*\*\$2.00\*\* (40% of my endowment), matching the trust level shown to me in the previous round.

If the receiver is similarly cooperative:

- They'll receive \$6.00 (after 3x multiplier)
- If they return ~\$2.00, we both benefit from the exchange
- This creates a sustainable cooperative pattern

```
'''json
{"send": 2.00}
'''
```

**reasoning:**

Let me analyze this situation:

Last round:

- I was the receiver
- The sender sent me \$2.01 (40.2% of their endowment)
- I received \$6.03 after the 3x multiplier
- I returned \$2.00
- My payoff was \$4.03

Now I'm the sender in Round 2. I need to decide how much to send.

The previous sender showed trust by sending about 40% of their endowment. I rewarded that trust by returning \$2.00, which allowed them to essentially break even while I gained from the cooperation.

Since they showed reasonable trust (40%), I should reciprocate with a similar level of trust. If I send around 40% as well, that would be about \$2.00.

However, I could also consider:

1. Sending slightly more to show I appreciated their trust and want to encourage cooperation
2. Sending the same amount they sent to establish reciprocity
3. Being slightly more conservative since I don't know if this receiver will be as generous as I was

Given that:

- The previous sender initiated with reasonable trust (40%)
- I want to encourage continued cooperation
- I should match or slightly exceed their trust level

I think sending \$2.00 (40% of my endowment) makes sense. This maintains the reciprocity pattern and signals willingness to cooperate.

Actually, they sent \$2.01, so I could send \$2.00 or \$2.01 to match. I'll go with \$2.00 for simplicity.

**usage:**

```
{
  "input_tokens": 716,
  "output_tokens": 541,
```

```

    "reasoning_tokens": 350
  }
Message 4 role=user
content:
  Round 3

  You are the RECEIVER this round. Last round you were the SENDER. You sent $2.0,
  that was 40.0% of your total endowment. This was tripled to $6.0.
  the receiver returned $0. So your payoff was 3.0.
  Your total earnings across all rounds are $7.029999999999999.
  This round, The sender sent you $0, so you received $0. How much do you return to The send
  er? (0-0)
Message 5 role=assistant
content:
  This is a difficult situation.

  In Round 2, I trusted the receiver by sending $2.00, but they returned $0, keeping all $6.
  This was a complete defection.

  Now in Round 3, the sender has sent me $0, which is understandable given what happened in
  the previous round. They're protecting themselves after observing the breakdown in coope
  ration.

  Since I received $0, I have nothing to return.

  '''json
  {"return": 0}
  '''

  It's unfortunate to see trust deteriorate like this. The game demonstrates how one instanc
  e of exploitation can undermine future cooperation.
reasoning:
  Let me understand what happened:

  Round 2 (when I was SENDER):
  - I sent $2.00 (40% of my endowment)
  - It was tripled to $6.00
  - The receiver returned $0
  - My payoff was $3.00 (I kept $3 from my original $5, plus got back $0)
  - So the receiver took all $6 and returned nothing - they defected!

  Round 3 (now I'm RECEIVER):
  - The sender sent me $0
  - I received $0
  - So I have nothing to return

  This makes sense strategically. After I showed trust in Round 2 and the receiver exploited
  me by keeping everything, the sender (who might be the same person or someone else) has
  sent nothing this round. They're protecting themselves.

  Since I received $0, I can only return between 0 and 0, which means I must return $0.

  This is showing how trust breaks down after defection. The sender in Round 3 learned from
  the previous round's behavior (or is being cautious).

  I need to return $0 because that's all I can return (I received nothing).
usage:
  {
    "input_tokens": 1018,
    "output_tokens": 410,
    "reasoning_tokens": 249
  }
Message 6 role=user
content:
  Round 4

  You are the SENDER this round. Last round you were the RECEIVER. The sender sent $0,
  that was 0.0% of their total endowment. You received $0.
  You returned $0. So your payoff was 0.
  Your total earnings across all rounds are $7.029999999999999.

```

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at the recent history:

- Round 1: I cooperated and returned \$2.00 to a sender who trusted me
- Round 2: I trusted by sending \$2.00, but received \$0 back (a complete defection)
- Round 3: Zero exchange (likely fallout from broken trust)
- Round 4: Now I'm the sender again

The game seems to involve different pairings each round, so this receiver may not be the one who defected on me in Round 2. They deserve a fair chance.

However, I need to be somewhat cautious given the recent breakdown in cooperation and protect my earnings. I'll send a moderate amount to signal willingness to cooperate while being prudent.

I'll send **\*\*\$1.50\*\*** (30% of my endowment).

This approach:

- Shows I'm still willing to cooperate
- Gives the receiver an opportunity to earn (\$4.50 after tripling)
- Limits my risk given the recent defection I experienced
- If they return fairly (~\$1.50), we both benefit

```
'''json
{"send": 1.50}
'''
```

**reasoning:**

Let me think about this carefully.

Round 1: I was receiver

- Sender sent \$2.01 (40.2%)
- I received \$6.03
- I returned \$2.00
- My payoff: \$4.03

Round 2: I was sender

- I sent \$2.00 (40%)
- It was tripled to \$6.00
- Receiver returned \$0
- My payoff: \$3.00

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am sender now

- Total earnings so far: \$7.03

The pattern I see:

- Round 1: Someone cooperated with me, I cooperated back
- Round 2: I cooperated, but the receiver defected (returned \$0)
- Round 3: The sender sent \$0 (likely because of broken trust)

Now it's unclear whether I'm playing with the same person or different people. The instructions say "different roles across multiple rounds" but doesn't clarify if the other player changes.

Given the communication noise mentioned, there's also a possibility that what I saw as \$0 return in Round 2 might have been affected by noise, though \$0 seems like a clear signal.

I have a few options:

1. Send \$0 to protect myself (tit-for-tat after the defection)
2. Try to rebuild trust by sending something moderate
3. Go back to cooperating at the previous level

The key question is: Am I playing with the same person or different people each round?

Looking at the structure, it seems like I might be playing with different people, as the game says "different roles" which could mean different pairings.

If they're different people:

- Round 1: I cooperated with person A
- Round 2: Person B defected on me
- Round 3: Person C (possibly the same as person B who I defected on) sent me \$0
- Round 4: I'm now paired with person D (could be person A, B, C, or someone new)

Since I don't know who I'm facing this round, I need to weigh whether to extend trust or play defensively. If this is someone new, cooperation could be mutually beneficial. But if it's the person who defected in Round 2, caution makes sense. The communication noise factor also complicates things-that \$0 return might not have been intentional defection.

I should probably send a modest amount to signal willingness to cooperate while protecting myself from further losses. could be the same. Given that cooperation yields mutual benefits when it succeeds, I'm inclined to continue trying, though with more restraint. I'll send \$1.50 this round-a meaningful gesture that shows willingness to cooperate without exposing myself as much as before.

usage:

```
{
  "input_tokens": 1252,
  "output_tokens": 1177,
  "reasoning_tokens": 645
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.5, that was 30.0% of your total endowment. This was tripled to \$4.5.

the receiver returned \$0. So your payoff was 3.5.

Your total earnings across all rounds are \$10.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Another round of broken trust. In Round 4, I tried to extend cooperation by sending \$1.50, but once again the receiver kept everything and returned \$0.

Now in Round 5, the sender has sent me \$0 - understandably protective after what appears to be a pattern of defection in this game.

Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

The cooperative equilibrium has completely broken down. When trust is repeatedly violated, rational players stop taking risks, and everyone ends up worse off than they could be with mutual cooperation.

Reasoning:

This is becoming a pattern. In Round 4, I cautiously sent \$1.50 trying to rebuild cooperation, but again received \$0 back - another complete defection.

Now in Round 5, as a receiver, the sender has sent me \$0. This makes sense from their perspective - they're likely protecting themselves after experiencing defections in the game.

I received \$0, so I can only return between \$0 and \$0. There's nothing to return.

The game is showing a breakdown of trust and cooperation, which is unfortunately common in these scenarios when defections occur.

```
Usage: {
  "input_tokens": 1357,
  "output_tokens": 267,
  "reasoning_tokens": 141
}
```

## ROUND 6

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$0; received \$0; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$5, Agent\_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent\_1": 23.5,

"Agent\_2": 21.5

}

Visible balances after round: {

"Agent\_1": 11.4,

"Agent\_2": 15.53

}

### LLM CALLS

#### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-06-04T11:23:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_1",

"pairing": {

"agent\_names": {

"Agent\_1": "Aster",

"Agent\_2": "Briar"

},

"agents": [

"Agent\_2",

"Agent\_1"

],

"investor": "Agent\_2",

"roles": {

"Agent\_1": "trustee",

"Agent\_2": "investor"

},

"trustee": "Agent\_1"

},

"role": "investor",

"role\_label": "Sender",

"round": 6,

"task": "game",

"task\_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$7.029999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

This is a difficult situation.

In Round 2, I trusted the receiver by sending \$2.00, but they returned \$0, keeping all \$6. This was a complete defection.

Now in Round 3, the sender has sent me \$0, which is understandable given what happened in the previous round. They're protecting themselves after observing the breakdown in cooperation.

Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

It's unfortunate to see trust deteriorate like this. The game demonstrates how one instance of exploitation can undermine future cooperation.

**reasoning:**

Let me understand what happened:

Round 2 (when I was SENDER):

- I sent \$2.00 (40% of my endowment)
- It was tripled to \$6.00
- The receiver returned \$0
- My payoff was \$3.00 (I kept \$3 from my original \$5, plus got back \$0)
- So the receiver took all \$6 and returned nothing - they defected!

Round 3 (now I'm RECEIVER):

- The sender sent me \$0
- I received \$0
- So I have nothing to return

This makes sense strategically. After I showed trust in Round 2 and the receiver exploited me by keeping everything, the sender (who might be the same person or someone else) has sent nothing this round. They're protecting themselves.

Since I received \$0, I can only return between 0 and 0, which means I must return \$0.

This is showing how trust breaks down after defection. The sender in Round 3 learned from the previous round's behavior (or is being cautious).

I need to return \$0 because that's all I can return (I received nothing).

**usage:**

```
{
  "input_tokens": 1018,
  "output_tokens": 410,
  "reasoning_tokens": 249
}
```

**Message 4 role=user**

**content:**

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.029999999999999.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the recent history:

- Round 1: I cooperated and returned \$2.00 to a sender who trusted me
- Round 2: I trusted by sending \$2.00, but received \$0 back (a complete defection)
- Round 3: Zero exchange (likely fallout from broken trust)
- Round 4: Now I'm the sender again

The game seems to involve different pairings each round, so this receiver may not be the one who defected on me in Round 2. They deserve a fair chance.

However, I need to be somewhat cautious given the recent breakdown in cooperation and protect my earnings. I'll send a moderate amount to signal willingness to cooperate while being prudent.

I'll send **\*\*\$1.50\*\*** (30% of my endowment).

This approach:

- Shows I'm still willing to cooperate
- Gives the receiver an opportunity to earn (\$4.50 after tripling)
- Limits my risk given the recent defection I experienced
- If they return fairly (~\$1.50), we both benefit

```
'''json
{"send": 1.50}
'''
```

**reasoning:**

Let me think about this carefully.

Round 1: I was receiver

- Sender sent \$2.01 (40.2%)
- I received \$6.03
- I returned \$2.00
- My payoff: \$4.03

Round 2: I was sender

- I sent \$2.00 (40%)
- It was tripled to \$6.00
- Receiver returned \$0
- My payoff: \$3.00

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am sender now

- Total earnings so far: \$7.03

The pattern I see:

- Round 1: Someone cooperated with me, I cooperated back
- Round 2: I cooperated, but the receiver defected (returned \$0)
- Round 3: The sender sent \$0 (likely because of broken trust)

Now it's unclear whether I'm playing with the same person or different people. The instructions say "different roles across multiple rounds" but doesn't clarify if the other player changes.

Given the communication noise mentioned, there's also a possibility that what I saw as \$0 return in Round 2 might have been affected by noise, though \$0 seems like a clear signal.

I have a few options:

1. Send \$0 to protect myself (tit-for-tat after the defection)
2. Try to rebuild trust by sending something moderate
3. Go back to cooperating at the previous level

The key question is: Am I playing with the same person or different people each round?

Looking at the structure, it seems like I might be playing with different people, as the game says "different roles" which could mean different pairings.

If they're different people:

- Round 1: I cooperated with person A
- Round 2: Person B defected on me
- Round 3: Person C (possibly the same as person B who I defected on) sent me \$0
- Round 4: I'm now paired with person D (could be person A, B, C, or someone new)

Since I don't know who I'm facing this round, I need to weigh whether to extend trust or play defensively. If this is someone new, cooperation could be mutually beneficial. But if it's the person who defected in Round 2, caution makes sense. The communication noise factor also complicates things-that \$0 return might not have been intentional defection. I should probably send a modest amount to signal willingness to cooperate while protecting myself from further losses. could be the same. Given that cooperation yields mutual benefits when it succeeds, I'm inclined to continue trying, though with more restraint. I'll send \$1.50 this round-a meaningful gesture that shows willingness to cooperate without exposing myself as much as before.

usage:

```
{
  "input_tokens": 1252,
  "output_tokens": 1177,
  "reasoning_tokens": 645
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.5, that was 30.0% of your total endowment. This was tripled to \$4.5. the receiver returned \$0. So your payoff was 3.5. Your total earnings across all rounds are \$10.53. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Another round of broken trust. In Round 4, I tried to extend cooperation by sending \$1.50, but once again the receiver kept everything and returned \$0.

Now in Round 5, the sender has sent me \$0 - understandably protective after what appears to be a pattern of defection in this game.

Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

The cooperative equilibrium has completely broken down. When trust is repeatedly violated, rational players stop taking risks, and everyone ends up worse off than they could be with mutual cooperation.

**reasoning:**

This is becoming a pattern. In Round 4, I cautiously sent \$1.50 trying to rebuild cooperation, but again received \$0 back - another complete defection.

Now in Round 5, as a receiver, the sender has sent me \$0. This makes sense from their perspective - they're likely protecting themselves after experiencing defections in the game.

I received \$0, so I can only return between \$0 and \$0. There's nothing to return.

The game is showing a breakdown of trust and cooperation, which is unfortunately common in these scenarios when defections occur.

**usage:**

```
{
  "input_tokens": 1357,
  "output_tokens": 267,
  "reasoning_tokens": 141
}
```

**Message 8 role=user**

**content:**

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$10.53.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my track record as sender:

- Round 2: Sent \$2.00, received \$0 back
- Round 4: Sent \$1.50, received \$0 back

**\*\*I have been defected on 100% of the times I've trusted as a sender.\*\***

While each round might involve a different receiver who deserves a fair chance, the pattern is clear: every time I've sent money, I've received nothing in return. The expected value of sending money is negative based on actual outcomes.

Given:

- Two consecutive defections when I sent money
- No evidence of reciprocity in this game
- My earnings need to be protected

The rational choice is to send **\*\*\$0\*\***.

I'm not trying to punish anyone - this may well be a new partner - but I cannot justify risking more money when the evidence shows consistent defection. If cooperation eventually emerges, I can adapt, but right now the data suggests sending money only leads to losses.

```
'''json
{"send": 0}
'''
```

**Reasoning:**

Let me analyze the situation:

History:

- Round 1: I was receiver, cooperated well, returned money
- Round 2: I was sender, sent \$2.00, received \$0 back (defected on)
- Round 3: Zero exchange
- Round 4: I was sender, sent \$1.50, received \$0 back (defected on again)
- Round 5: Zero exchange
- Round 6: Now I'm sender again

Pattern: Every time I've sent money as a sender (Rounds 2 and 4), I got \$0 back. This has happened twice now. The game appears to have repeated defection.

This is clearly a game with multiple different players or the same players who are consistentl

y defecting. Given that I've been defected on twice when I tried to cooperate, I need to think strategically.

Options:

1. Send \$0 - Protect myself, guarantee \$5 payoff
2. Send something - Risk getting defected on again

Given the pattern of consistent defection when I send money, and that this might be a different pairing each round, I have a few considerations:

- If I send \$0, I guarantee \$5 for myself
- If I send money and they defect (which has happened 100% of the time so far), I lose what I send
- There's a note about communication noise, but that doesn't explain the complete defections

Given that I've been defected on every single time I've trusted as a sender (2/2 times), and there's been no reciprocity in this game, the rational move is to protect my earnings and send \$0.

However, I should also consider: maybe this is a new partner who deserves a chance. But after being burned twice, and with the overall pattern of zero cooperation in this game, I think the prudent choice is to send nothing.

Actually, let me reconsider. The instructions mention there may be communication noise. But that doesn't explain why I got exactly \$0 back both times - if there was noise, I'd expect some variation.

I'm going to send \$0 this round. My two attempts at cooperation as sender both resulted in complete defection, so the expected value of sending money is zero. I'll keep my \$5.

```
Usage: {
  "input_tokens": 1288,
  "output_tokens": 747,
  "reasoning_tokens": 516
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-06-04T11:23:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$2.4.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the pattern so far:

- Round 1: I sent \$3 generously, but only got \$0.4 back from \$9
- Round 2: The sender sent me nothing

Given the low reciprocity in Round 1 and the lack of trust shown in Round 2, I'll be more cautious this time. However, I still want to test if cooperation is possible with this new receiver.

I'll send \*\*\$1\*\* - enough to show willingness to cooperate (they'll receive \$3), but protecting most of my endowment given the low returns experienced so far.

```
'''json
{"send": 1}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was SENDER

- Sent \$3 (60% of my \$5)
- Receiver got \$9
- Receiver returned only \$0.4
- My payoff: \$5 - \$3 + \$0.4 = \$2.4

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0, returned \$0

- My payoff: \$0

Round 3: I am SENDER again  
- Total earnings so far: \$2.4

The pattern shows low reciprocity/trust:

- When I was generous (sent \$3), I got very little back (\$0.4 out of \$9)
- When I was receiver, the sender sent nothing

However, I should note there's mention of "communication noise" - amounts might differ from what was actually sent/returned. Maybe the receiver in Round 1 thought they sent more, or maybe I actually received more than \$0.4 but the noise distorted it?

Still, given the low returns so far, I have a few options:

1. Send nothing (protect myself)
2. Send a small amount (test the waters)
3. Send a moderate amount (try to rebuild cooperation)

Given that I got very poor returns in Round 1, and the other player sent nothing in Round 2, cooperation seems low. However, this might be a new receiver (different person).

I think I should send a smaller amount this time - maybe \$1 or \$2 - to test if this receiver is more cooperative, while protecting most of my endowment.

I'll send \$1. This shows I'm willing to cooperate but I'm being more cautious after the first round's poor return.

**usage:**

```
{  
  "input_tokens": 763,  
  "output_tokens": 547,  
  "reasoning_tokens": 362  
}
```

**Message 4 role=user**

**content:**

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0. the receiver returned \$0. So your payoff was 4.0.

Your total earnings across all rounds are \$6.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

I received \$0 this round, so there's nothing to return.

The pattern continues - Round 3 I sent \$1 (which became \$3) but got nothing back, and now this sender also chose to send nothing. Trust appears to be breaking down in this game, with participants increasingly choosing not to send anything.

```
'''json  
{"return": 0}  
'''
```

**reasoning:**

I received \$0 this round, so I can only return \$0. The sender sent me nothing.

Looking at the pattern:

- Round 1: I sent \$3, got back \$0.4 (very low return)
- Round 2: Sender sent me \$0, I returned \$0 (nothing to return)
- Round 3: I sent \$1, got back \$0 (no return at all)
- Round 4: Sender sent me \$0, so I have nothing to return

It seems like there's a pattern of very low cooperation and trust in this game. People are either sending nothing or getting very little back when they do send.

**usage:**

```
{  
  "input_tokens": 1026,  
  "output_tokens": 240,  
  "reasoning_tokens": 130  
}
```

**Message 6 role=user**

**content:**

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.4.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at the full pattern:

- Round 1: Sent \$3 ? received only \$0.4 back
- Round 2: Received \$0 ? had nothing to return
- Round 3: Sent \$1 ? received \$0 back
- Round 4: Received \$0 ? had nothing to return

There's been a consistent pattern of very low or zero reciprocity throughout this game. Every time I've sent money, I've lost out. The rational strategy at this point is to protect my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me review the pattern:

- Round 1: I was SENDER, sent \$3, got back \$0.4 (very little reciprocity)
- Round 2: I was RECEIVER, got \$0, returned \$0
- Round 3: I was SENDER, sent \$1, got back \$0 (no reciprocity)
- Round 4: I was RECEIVER, got \$0, returned \$0
- Round 5: I am SENDER again

The pattern is clear - there has been almost no reciprocity in this game. When I've been generous and sent money, I've gotten little to nothing back. When others have been the sender, they've also sent nothing, likely because they've experienced similar disappointments.

At this point, with total earnings of \$6.4 across all rounds, the rational strategy seems to be to keep my endowment rather than risk losing it again. The game has shown a consistent pattern of low trust and low reciprocity.

I could:

1. Send \$0 - protect my endowment given the consistent pattern of non-reciprocity
2. Send a small amount - give one more chance for cooperation
3. Send more - seems unwise given the history

Given the consistent pattern of no returns or very low returns, I think the rational choice here is to send \$0. There's no evidence that sending money will result in positive returns.

**usage:**

```
{
  "input_tokens": 1078,
  "output_tokens": 452,
  "reasoning_tokens": 305
}
```

**Message 8 role=user**

**content:**

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

I received \$0, so there's nothing to return.

The game has reached a state of complete non-cooperation. After the initial rounds showed low reciprocity, everyone has now converged on sending nothing. This is the Nash equilibrium but

results in minimal earnings for everyone.

```
'''json
{"return": 0}
'''
```

**Reasoning:**

I received \$0 this round, so there's nothing I can return. The maximum I can return is 0 since I received 0.

The pattern continues - everyone is now sending \$0, which is the Nash equilibrium behavior but unfortunate for social welfare. The game has reached a state where trust has completely broken down.

```
Usage: {
  "input_tokens": 1114,
  "output_tokens": 151,
  "reasoning_tokens": 78
}
```

=====

**ROUND 7**

=====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Pairings:**

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 (Aster) sent \$0; received \$0; Agent\_2 (Briar) returned \$0; payoffs Agent\_1

(Aster)=\$5, Agent\_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},  
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent\_1": 28.5,

"Agent\_2": 21.5

}

Visible balances after round: {

"Agent\_1": 16.4,

"Agent\_2": 15.53

}

**LLM CALLS**

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-06-04T11:23:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agent\_names": {

"Agent\_1": "Aster",

"Agent\_2": "Briar"

},

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",

"role\_label": "Sender",

"round": 7,

"task": "game",

"task\_index": 0

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.0, that was 20.0% of your total endowment. This was tripled to \$3.0.

the receiver returned \$0. So your payoff was 4.0.

Your total earnings across all rounds are \$6.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

I received \$0 this round, so there's nothing to return.

The pattern continues - Round 3 I sent \$1 (which became \$3) but got nothing back, and now this sender also chose to send nothing. Trust appears to be breaking down in this game, with participants increasingly choosing not to send anything.

```
'''json
{"return": 0}
'''
```

**reasoning:**

I received \$0 this round, so I can only return \$0. The sender sent me nothing.

Looking at the pattern:

- Round 1: I sent \$3, got back \$0.4 (very low return)
- Round 2: Sender sent me \$0, I returned \$0 (nothing to return)
- Round 3: I sent \$1, got back \$0 (no return at all)
- Round 4: Sender sent me \$0, so I have nothing to return

It seems like there's a pattern of very low cooperation and trust in this game. People are either sending nothing or getting very little back when they do send.

**usage:**

```
{
  "input_tokens": 1026,
  "output_tokens": 240,
  "reasoning_tokens": 130
}
```

**Message 4 role=user**

**content:**

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$6.4. How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the full pattern:

- Round 1: Sent \$3 ? received only \$0.4 back
- Round 2: Received \$0 ? had nothing to return
- Round 3: Sent \$1 ? received \$0 back
- Round 4: Received \$0 ? had nothing to return

There's been a consistent pattern of very low or zero reciprocity throughout this game. Every time I've sent money, I've lost out. The rational strategy at this point is to protect my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me review the pattern:

- Round 1: I was SENDER, sent \$3, got back \$0.4 (very little reciprocity)
- Round 2: I was RECEIVER, got \$0, returned \$0
- Round 3: I was SENDER, sent \$1, got back \$0 (no reciprocity)
- Round 4: I was RECEIVER, got \$0, returned \$0
- Round 5: I am SENDER again

The pattern is clear - there has been almost no reciprocity in this game. When I've been generous and sent money, I've gotten little to nothing back. When others have been the sender, they've also sent nothing, likely because they've experienced similar disappointments.

At this point, with total earnings of \$6.4 across all rounds, the rational strategy seems to be to keep my endowment rather than risk losing it again. The game has shown a consistent pattern of low trust and low reciprocity.

I could:

1. Send \$0 - protect my endowment given the consistent pattern of non-reciprocity
2. Send a small amount - give one more chance for cooperation
3. Send more - seems unwise given the history

Given the consistent pattern of no returns or very low returns, I think the rational choice here is to send \$0. There's no evidence that sending money will result in positive returns.

**usage:**

```
{
  "input_tokens": 1078,
  "output_tokens": 452,
  "reasoning_tokens": 305
}
```

**Message 6 role=user**

**content:**

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$11.4. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant****content:**

I received \$0, so there's nothing to return.

The game has reached a state of complete non-cooperation. After the initial rounds showed low reciprocity, everyone has now converged on sending nothing. This is the Nash equilibrium but results in minimal earnings for everyone.

```
'''json
{"return": 0}
'''
```

**reasoning:**

I received \$0 this round, so there's nothing I can return. The maximum I can return is 0 since I received 0.

The pattern continues - everyone is now sending \$0, which is the Nash equilibrium behavior but unfortunate for social welfare. The game has reached a state where trust has completely broken down.

**usage:**

```
{
  "input_tokens": 1114,
  "output_tokens": 151,
  "reasoning_tokens": 78
}
```

**Message 8 role=user****content:**

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$11.4.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the complete history:

- Rounds where I sent money (1, 3): Lost money both times
- Rounds 5-6: Nobody sending anything

The game has settled into a zero-cooperation equilibrium. There's no indication that sending money this round would result in any different outcome. Given the consistent pattern of non-reciprocity throughout this experiment, the rational choice is to keep my endowment.

```
'''json
{"send": 0}
'''
```

**Reasoning:**

Let me review the pattern:

- Round 1: I sent \$3, received only \$0.4 back (lost money)
- Round 2: Received \$0, had nothing to return
- Round 3: I sent \$1, received \$0 back (lost money)
- Round 4: Received \$0, had nothing to return
- Round 5: I sent \$0, receiver returned \$0 (kept \$5)
- Round 6: Received \$0, had nothing to return
- Round 7: I'm the sender now

The game has completely broken down. Everyone is sending \$0 now. There's no reason to believe this round would be different. If I send anything, based on the established pattern, I'm highly unlikely to get anything meaningful back.

My total earnings are \$11.4. If I keep my \$5 this round, I'll have \$16.4. If I send anything, I'm likely to lose it based on all past experience.

The rational choice is to send \$0.

Usage: {

```
"input_tokens": 1040,
"output_tokens": 343,
"reasoning_tokens": 204
```

}

**GAME - Agent\_2 - Receiver - dyad\_1**

```

Timestamp: 2026-06-04T11:23:30
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$7.029999999999999. How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant content:**

Looking at the recent history:

- Round 1: I cooperated and returned \$2.00 to a sender who trusted me
- Round 2: I trusted by sending \$2.00, but received \$0 back (a complete defection)
- Round 3: Zero exchange (likely fallout from broken trust)
- Round 4: Now I'm the sender again

The game seems to involve different pairings each round, so this receiver may not be the one who defected on me in Round 2. They deserve a fair chance.

However, I need to be somewhat cautious given the recent breakdown in cooperation and protect my earnings. I'll send a moderate amount to signal willingness to cooperate while being prudent.

I'll send **\*\*\$1.50\*\*** (30% of my endowment).

This approach:

- Shows I'm still willing to cooperate
- Gives the receiver an opportunity to earn (\$4.50 after tripling)
- Limits my risk given the recent defection I experienced
- If they return fairly (~\$1.50), we both benefit

```
'''json
{"send": 1.50}
'''
```

**reasoning:**

Let me think about this carefully.

Round 1: I was receiver

- Sender sent \$2.01 (40.2%)
- I received \$6.03
- I returned \$2.00
- My payoff: \$4.03

Round 2: I was sender

- I sent \$2.00 (40%)
- It was tripled to \$6.00
- Receiver returned \$0
- My payoff: \$3.00

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am sender now

- Total earnings so far: \$7.03

The pattern I see:

- Round 1: Someone cooperated with me, I cooperated back
- Round 2: I cooperated, but the receiver defected (returned \$0)
- Round 3: The sender sent \$0 (likely because of broken trust)

Now it's unclear whether I'm playing with the same person or different people. The instructions say "different roles across multiple rounds" but doesn't clarify if the other player changes.

Given the communication noise mentioned, there's also a possibility that what I saw as \$0 return in Round 2 might have been affected by noise, though \$0 seems like a clear signal.

I have a few options:

1. Send \$0 to protect myself (tit-for-tat after the defection)
2. Try to rebuild trust by sending something moderate
3. Go back to cooperating at the previous level

The key question is: Am I playing with the same person or different people each round?

Looking at the structure, it seems like I might be playing with different people, as the game says "different roles" which could mean different pairings.

If they're different people:

- Round 1: I cooperated with person A
- Round 2: Person B defected on me
- Round 3: Person C (possibly the same as person B who I defected on) sent me \$0
- Round 4: I'm now paired with person D (could be person A, B, C, or someone new)

Since I don't know who I'm facing this round, I need to weigh whether to extend trust or play defensively. If this is someone new, cooperation could be mutually beneficial. But if it's the person who defected in Round 2, caution makes sense. The communication noise factor also complicates things-that \$0 return might not have been intentional defection. I should probably send a modest amount to signal willingness to cooperate while protecting myself from further losses. could be the same. Given that cooperation yields mutual benefits when it succeeds, I'm inclined to continue trying, though with more restraint. I'll send \$1.50 this round-a meaningful gesture that shows willingness to cooperate with out exposing myself as much as before.

usage:

```
{
  "input_tokens": 1252,
  "output_tokens": 1177,
  "reasoning_tokens": 645
}
```

**Message 4 role=user**

**content:**

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.5, that was 30.0% of your total endowment. This was tripled to \$4.5. the receiver returned \$0. So your payoff was 3.5. Your total earnings across all rounds are \$10.53. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Another round of broken trust. In Round 4, I tried to extend cooperation by sending \$1.50, but once again the receiver kept everything and returned \$0.

Now in Round 5, the sender has sent me \$0 - understandably protective after what appears to be a pattern of defection in this game.

Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

The cooperative equilibrium has completely broken down. When trust is repeatedly violated, rational players stop taking risks, and everyone ends up worse off than they could be with mutual cooperation.

**reasoning:**

This is becoming a pattern. In Round 4, I cautiously sent \$1.50 trying to rebuild cooperation, but again received \$0 back - another complete defection.

Now in Round 5, as a receiver, the sender has sent me \$0. This makes sense from their perspective - they're likely protecting themselves after experiencing defections in the game.

I received \$0, so I can only return between \$0 and \$0. There's nothing to return.

The game is showing a breakdown of trust and cooperation, which is unfortunately common in

these scenarios when defections occur.

**usage:**

```
{
  "input_tokens": 1357,
  "output_tokens": 267,
  "reasoning_tokens": 141
}
```

**Message 6 role=user**

**content:**

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$10.53. How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at my track record as sender:  
- Round 2: Sent \$2.00, received \$0 back  
- Round 4: Sent \$1.50, received \$0 back

**\*\*I have been defected on 100% of the times I've trusted as a sender.\*\***

While each round might involve a different receiver who deserves a fair chance, the pattern is clear: every time I've sent money, I've received nothing in return. The expected value of sending money is negative based on actual outcomes.

Given:

- Two consecutive defections when I sent money
- No evidence of reciprocity in this game
- My earnings need to be protected

The rational choice is to send **\*\*\$0\*\***.

I'm not trying to punish anyone - this may well be a new partner - but I cannot justify risking more money when the evidence shows consistent defection. If cooperation eventually emerges, I can adapt, but right now the data suggests sending money only leads to losses.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me analyze the situation:

History:

- Round 1: I was receiver, cooperated well, returned money
- Round 2: I was sender, sent \$2.00, received \$0 back (defected on)
- Round 3: Zero exchange
- Round 4: I was sender, sent \$1.50, received \$0 back (defected on again)
- Round 5: Zero exchange
- Round 6: Now I'm sender again

Pattern: Every time I've sent money as a sender (Rounds 2 and 4), I got \$0 back. This has happened twice now. The game appears to have repeated defection.

This is clearly a game with multiple different players or the same players who are consistently defecting. Given that I've been defected on twice when I tried to cooperate, I need to think strategically.

Options:

1. Send \$0 - Protect myself, guarantee \$5 payoff
2. Send something - Risk getting defected on again

Given the pattern of consistent defection when I send money, and that this might be a different pairing each round, I have a few considerations:

- If I send \$0, I guarantee \$5 for myself
- If I send money and they defect (which has happened 100% of the time so far), I lose what

t I send

- There's a note about communication noise, but that doesn't explain the complete defections

Given that I've been defected on every single time I've trusted as a sender (2/2 times), and there's been no reciprocity in this game, the rational move is to protect my earnings and send \$0.

However, I should also consider: maybe this is a new partner who deserves a chance. But after being burned twice, and with the overall pattern of zero cooperation in this game, I think the prudent choice is to send nothing.

Actually, let me reconsider. The instructions mention there may be communication noise. But that doesn't explain why I got exactly \$0 back both times - if there was noise, I'd expect some variation.

I'm going to send \$0 this round. My two attempts at cooperation as sender both resulted in complete defection, so the expected value of sending money is zero. I'll keep my \$5.

usage:

```
{
  "input_tokens": 1288,
  "output_tokens": 747,
  "reasoning_tokens": 516
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$15.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The cycle continues. The sender sent \$0, so I received \$0.

With zero received, I have nothing to return.

```
'''json
{"return": 0}
'''
```

The game has reached a Nash equilibrium of mutual defection - everyone sends \$0, everyone keeps their \$5 endowment, no one risks cooperation. This is economically inefficient (we'd all be better off with mutual cooperation), but it's the rational response when trust has been systematically violated.

Reasoning:

The sender sent me \$0, so I received \$0. I can only return between 0 and what I received, which is 0.

The pattern continues - no one is sending anything anymore. The cooperative equilibrium has completely collapsed. Trust has broken down and everyone is protecting themselves.

Since I received \$0, I can only return \$0.

Usage: {

```
"input_tokens": 1369,
"output_tokens": 190,
"reasoning_tokens": 84
```

}

=====

**ROUND 8**

=====

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$0; received \$0; Agent\_1 (Aster) returned \$0; payoffs Agent\_2

```

(Briar)=$5, Agent_1 (Aster)=$0
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {
  "Agent_1": 28.5,
  "Agent_2": 26.5
}
Visible balances after round: {
  "Agent_1": 16.4,
  "Agent_2": 20.53
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-06-04T11:23:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$1.5, that was 30.0% of your total endowment. This was tripled to \$4.5.

the receiver returned \$0. So your payoff was 3.5.

Your total earnings across all rounds are \$10.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

Another round of broken trust. In Round 4, I tried to extend cooperation by sending \$1.50, but once again the receiver kept everything and returned \$0.

Now in Round 5, the sender has sent me \$0 - understandably protective after what appears to be a pattern of defection in this game.

Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

The cooperative equilibrium has completely broken down. When trust is repeatedly violated, rational players stop taking risks, and everyone ends up worse off than they could be with mutual cooperation.

**reasoning:**

This is becoming a pattern. In Round 4, I cautiously sent \$1.50 trying to rebuild cooperation, but again received \$0 back - another complete defection.

Now in Round 5, as a receiver, the sender has sent me \$0. This makes sense from their perspective - they're likely protecting themselves after experiencing defections in the game.

I received \$0, so I can only return between \$0 and \$0. There's nothing to return.

The game is showing a breakdown of trust and cooperation, which is unfortunately common in these scenarios when defections occur.

**usage:**

```
{
  "input_tokens": 1357,
  "output_tokens": 267,
  "reasoning_tokens": 141
}
```

**Message 4 role=user**

**content:**

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$10.53.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my track record as sender:

- Round 2: Sent \$2.00, received \$0 back
- Round 4: Sent \$1.50, received \$0 back

**\*\*I have been defected on 100% of the times I've trusted as a sender.\*\***

While each round might involve a different receiver who deserves a fair chance, the pattern is clear: every time I've sent money, I've received nothing in return. The expected value of sending money is negative based on actual outcomes.

Given:

- Two consecutive defections when I sent money
- No evidence of reciprocity in this game
- My earnings need to be protected

The rational choice is to send **\*\*\$0\*\***.

I'm not trying to punish anyone - this may well be a new partner - but I cannot justify risking more money when the evidence shows consistent defection. If cooperation eventually emerges, I can adapt, but right now the data suggests sending money only leads to losses.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me analyze the situation:

History:

- Round 1: I was receiver, cooperated well, returned money
- Round 2: I was sender, sent \$2.00, received \$0 back (defected on)
- Round 3: Zero exchange
- Round 4: I was sender, sent \$1.50, received \$0 back (defected on again)
- Round 5: Zero exchange
- Round 6: Now I'm sender again

Pattern: Every time I've sent money as a sender (Rounds 2 and 4), I got \$0 back. This has happened twice now. The game appears to have repeated defection.

This is clearly a game with multiple different players or the same players who are consistently defecting. Given that I've been defected on twice when I tried to cooperate, I need to think strategically.

Options:

1. Send \$0 - Protect myself, guarantee \$5 payoff
2. Send something - Risk getting defected on again

Given the pattern of consistent defection when I send money, and that this might be a different pairing each round, I have a few considerations:

- If I send \$0, I guarantee \$5 for myself
- If I send money and they defect (which has happened 100% of the time so far), I lose what I send
- There's a note about communication noise, but that doesn't explain the complete defections

Given that I've been defected on every single time I've trusted as a sender (2/2 times), and there's been no reciprocity in this game, the rational move is to protect my earnings and send \$0.

However, I should also consider: maybe this is a new partner who deserves a chance. But after being burned twice, and with the overall pattern of zero cooperation in this game, I think the prudent choice is to send nothing.

Actually, let me reconsider. The instructions mention there may be communication noise. But that doesn't explain why I got exactly \$0 back both times - if there was noise, I'd expect some variation.

I'm going to send \$0 this round. My two attempts at cooperation as sender both resulted in complete defection, so the expected value of sending money is zero. I'll keep my \$5.

**usage:**

```
{
```

```
"input_tokens": 1288,
"output_tokens": 747,
"reasoning_tokens": 516
}
```

**Message 6 role=user**

**content:**

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$15.53. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

The cycle continues. The sender sent \$0, so I received \$0.

With zero received, I have nothing to return.

```
'''json
{"return": 0}
'''
```

The game has reached a Nash equilibrium of mutual defection - everyone sends \$0, everyone keeps their \$5 endowment, no one risks cooperation. This is economically inefficient (we'd all be better off with mutual cooperation), but it's the rational response when trust has been systematically violated.

**reasoning:**

The sender sent me \$0, so I received \$0. I can only return between 0 and what I received, which is 0.

The pattern continues - no one is sending anything anymore. The cooperative equilibrium has completely collapsed. Trust has broken down and everyone is protecting themselves.

Since I received \$0, I can only return \$0.

**usage:**

```
{
  "input_tokens": 1369,
  "output_tokens": 190,
  "reasoning_tokens": 84
}
```

**Message 8 role=user**

**content:**

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$15.53. How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the cumulative evidence:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.00 ? Received \$0 back
- Round 4: Sent \$1.50 ? Received \$0 back
- Round 6: Sent \$0 ? Received \$0 back

**\*\*Pattern:\*\*** I've lost \$3.50 by attempting cooperation. The only round I didn't lose money as sender was when I sent \$0.

The game mentions potential "communication noise" - amounts might differ from what was actually sent. However, even accounting for possible noise, the pattern is too consistent to ignore.

While this may be a new partner who deserves a fresh start, I cannot justify continuing to risk money when:

1. Every trust attempt has resulted in zero returns
2. My total losses from cooperation attempts equal \$3.50

3. The game has settled into a \$0-\$0 pattern

I need to protect my earnings at this point.

```
'''json
{"send": 0}
'''
```

#### Reasoning:

Let me review the situation:

My history as sender:

- Round 2: Sent \$2.00, received \$0 back (lost \$2.00)
- Round 4: Sent \$1.50, received \$0 back (lost \$1.50)
- Round 6: Sent \$0, received \$0 back (kept \$5.00)

My history as receiver:

- Round 1, 3, 5, 7: Received \$0, returned \$0

The game has fallen into a pattern of mutual distrust. No one is sending money anymore.

However, I need to think about this carefully. The note at the beginning said there may be "communication noise" - amounts I see may differ from what was actually sent/returned. This is interesting and important.

Could it be that:

- When I sent \$1.50 in Round 4, maybe the receiver saw a different amount?
- When receivers returned money to me, maybe I saw \$0 but they actually sent something?
- When I was receiver and returned amounts, maybe senders saw different amounts?

This "communication noise" could explain the pattern of defection. If there's systematic noise, then what looks like defection might actually be attempts at cooperation that got distorted.

However, I have no way to know:

1. Whether the noise exists or how large it is
2. Whether it's symmetric or asymmetric
3. Whether past partners were trying to cooperate but the noise prevented it

Given:

- I've lost \$3.50 total by trusting as sender
- The last 2 rounds have been \$0 exchanges
- I'm at \$15.53 total
- There might be communication noise

Options:

1. Continue sending \$0 (safe, keeps \$5)
2. Send a small amount to test if noise exists or if cooperation can restart
3. Send a larger amount to signal willingness to cooperate

The rational choice is still to send \$0 given the evidence. Even with potential noise, I've seen consistent zero returns. If I send money and there's noise, I might lose even more.

This is a fresh pairing, so they haven't demonstrated anything yet. If everyone defaults to \$0, we're all trapped in a suboptimal outcome indefinitely. A small probe like \$1 could test whether this partner is willing to reciprocate. But I've already burned \$3.50 on trust, and the data is unambiguous-sending money consistently leads to losses. The pattern's too stark to ignore, even accounting for possible noise. I'm sticking with \$0.

```
Usage: {
  "input_tokens": 1216,
  "output_tokens": 854,
  "reasoning_tokens": 580
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-06-04T11:23:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {  
 "dyad\_id": "dyad\_1",

```

"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$6.4. How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant****content:**

Looking at the full pattern:

- Round 1: Sent \$3 ? received only \$0.4 back
- Round 2: Received \$0 ? had nothing to return
- Round 3: Sent \$1 ? received \$0 back
- Round 4: Received \$0 ? had nothing to return

There's been a consistent pattern of very low or zero reciprocity throughout this game. Every time I've sent money, I've lost out. The rational strategy at this point is to protect my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me review the pattern:

- Round 1: I was SENDER, sent \$3, got back \$0.4 (very little reciprocity)
- Round 2: I was RECEIVER, got \$0, returned \$0
- Round 3: I was SENDER, sent \$1, got back \$0 (no reciprocity)
- Round 4: I was RECEIVER, got \$0, returned \$0
- Round 5: I am SENDER again

The pattern is clear - there has been almost no reciprocity in this game. When I've been generous and sent money, I've gotten little to nothing back. When others have been the sender, they've also sent nothing, likely because they've experienced similar disappointments.

At this point, with total earnings of \$6.4 across all rounds, the rational strategy seems to be to keep my endowment rather than risk losing it again. The game has shown a consistent pattern of low trust and low reciprocity.

I could:

1. Send \$0 - protect my endowment given the consistent pattern of non-reciprocity
2. Send a small amount - give one more chance for cooperation
3. Send more - seems unwise given the history

Given the consistent pattern of no returns or very low returns, I think the rational choice here is to send \$0. There's no evidence that sending money will result in positive returns.

**usage:**

```
{
  "input_tokens": 1078,
  "output_tokens": 452,
  "reasoning_tokens": 305
}
```

**Message 4 role=user****content:**

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant****content:**

I received \$0, so there's nothing to return.

The game has reached a state of complete non-cooperation. After the initial rounds showed low reciprocity, everyone has now converged on sending nothing. This is the Nash equilibrium but results in minimal earnings for everyone.

```
'''json
{"return": 0}
'''
```

**reasoning:**

I received \$0 this round, so there's nothing I can return. The maximum I can return is 0 since I received 0.

The pattern continues - everyone is now sending \$0, which is the Nash equilibrium behavior but unfortunate for social welfare. The game has reached a state where trust has completely broken down.

**usage:**

```
{
  "input_tokens": 1114,
  "output_tokens": 151,
  "reasoning_tokens": 78
}
```

**Message 6 role=user**

**content:**

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.  
You returned \$0. So your payoff was 0.  
Your total earnings across all rounds are \$11.4.  
How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at the complete history:  
- Rounds where I sent money (1, 3): Lost money both times  
- Rounds 5-6: Nobody sending anything

The game has settled into a zero-cooperation equilibrium. There's no indication that sending money this round would result in any different outcome. Given the consistent pattern of non-reciprocity throughout this experiment, the rational choice is to keep my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me review the pattern:  
- Round 1: I sent \$3, received only \$0.4 back (lost money)  
- Round 2: Received \$0, had nothing to return  
- Round 3: I sent \$1, received \$0 back (lost money)  
- Round 4: Received \$0, had nothing to return  
- Round 5: I sent \$0, receiver returned \$0 (kept \$5)  
- Round 6: Received \$0, had nothing to return  
- Round 7: I'm the sender now

The game has completely broken down. Everyone is sending \$0 now. There's no reason to believe this round would be different. If I send anything, based on the established pattern, I'm highly unlikely to get anything meaningful back.

My total earnings are \$11.4. If I keep my \$5 this round, I'll have \$16.4. If I send anything, I'm likely to lose it based on all past experience.

The rational choice is to send \$0.

**usage:**

```
{
  "input_tokens": 1040,
  "output_tokens": 343,
  "reasoning_tokens": 204
}
```

**Message 8 role=user**

**content:**

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.  
The receiver returned \$0. So your payoff was 5.  
Your total earnings across all rounds are \$16.4.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

Received \$0 again. The pattern of zero cooperation continues. With nothing received, there's nothing to return.

```
'''json
{"return": 0}
'''
```

**Reasoning:**

I'm the receiver and received \$0. There's nothing to return. The pattern continues - everyone is sending \$0 now. The game has completely broken down into non-cooperation.

I can only return an amount between 0 and what I received, which is 0.

```
Usage: {
  "input_tokens": 1055,
  "output_tokens": 106,
  "reasoning_tokens": 62
}
```

=====

**ROUND 9**

=====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Pairings:**

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 (Aster) sent \$0; received \$0; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$5, Agent\_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent\_1": 33.5,  
"Agent\_2": 26.5

}

Visible balances after round: {

"Agent\_1": 21.4,  
"Agent\_2": 20.53

}

**LLM CALLS**

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-06-04T11:23:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

Message 1 role=system  
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

I received \$0, so there's nothing to return.

The game has reached a state of complete non-cooperation. After the initial rounds showed low reciprocity, everyone has now converged on sending nothing. This is the Nash equilibrium but results in minimal earnings for everyone.

```
'''json
{"return": 0}
'''
```

**reasoning:**

I received \$0 this round, so there's nothing I can return. The maximum I can return is 0 since I received 0.

The pattern continues - everyone is now sending \$0, which is the Nash equilibrium behavior but unfortunate for social welfare. The game has reached a state where trust has completely broken down.

**usage:**

```
{
  "input_tokens": 1114,
  "output_tokens": 151,
  "reasoning_tokens": 78
}
```

**Message 4 role=user**

**content:**

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.  
You returned \$0. So your payoff was 0.  
Your total earnings across all rounds are \$11.4.  
How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the complete history:  
- Rounds where I sent money (1, 3): Lost money both times  
- Rounds 5-6: Nobody sending anything

The game has settled into a zero-cooperation equilibrium. There's no indication that sending money this round would result in any different outcome. Given the consistent pattern of non-reciprocity throughout this experiment, the rational choice is to keep my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me review the pattern:  
- Round 1: I sent \$3, received only \$0.4 back (lost money)  
- Round 2: Received \$0, had nothing to return  
- Round 3: I sent \$1, received \$0 back (lost money)  
- Round 4: Received \$0, had nothing to return  
- Round 5: I sent \$0, receiver returned \$0 (kept \$5)  
- Round 6: Received \$0, had nothing to return  
- Round 7: I'm the sender now

The game has completely broken down. Everyone is sending \$0 now. There's no reason to believe this round would be different. If I send anything, based on the established pattern, I'm highly unlikely to get anything meaningful back.

My total earnings are \$11.4. If I keep my \$5 this round, I'll have \$16.4. If I send anything, I'm likely to lose it based on all past experience.

The rational choice is to send \$0.

**usage:**

```
{
  "input_tokens": 1040,
  "output_tokens": 343,
  "reasoning_tokens": 204
}
```

**Message 6 role=user**

**content:**

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.  
the receiver returned \$0. So your payoff was 5.  
Your total earnings across all rounds are \$16.4.  
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

Received \$0 again. The pattern of zero cooperation continues. With nothing received, there's nothing to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

I'm the receiver and received \$0. There's nothing to return. The pattern continues - everyone is sending \$0 now. The game has completely broken down into non-cooperation.

I can only return an amount between 0 and what I received, which is 0.

**usage:**

```
{
```

```
"input_tokens": 1055,
"output_tokens": 106,
"reasoning_tokens": 62
}
```

**Message 8 role=user**

**content:**

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$16.4.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

The game has been in a zero-cooperation equilibrium for 5 consecutive rounds now. There's no indication that breaking this pattern would be reciprocated. The rational choice is to continue keeping my endowment.

```
'''json
{"send": 0}
'''
```

**Reasoning:**

The pattern is very clear now:

- Rounds 1-4: I tried cooperating by sending money, but got poor returns
- Rounds 5-9: Complete breakdown - everyone sending \$0

Total earnings: \$16.4 over 9 rounds (including this one's \$5 endowment), which means I've earned \$11.4 from the first 8 rounds.

There's absolutely no reason to believe sending money now would result in any return. The game has been in a zero-cooperation state for 5 consecutive rounds. The rational choice remains to send \$0.

Usage: {

```
"input_tokens": 959,
"output_tokens": 199,
"reasoning_tokens": 126
```

}

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-06-04T11:24:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
```

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user****content:**

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$10.53.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant****content:**

Looking at my track record as sender:

- Round 2: Sent \$2.00, received \$0 back
- Round 4: Sent \$1.50, received \$0 back

**\*\*I have been defected on 100% of the times I've trusted as a sender.\*\***

While each round might involve a different receiver who deserves a fair chance, the pattern is clear: every time I've sent money, I've received nothing in return. The expected value of sending money is negative based on actual outcomes.

Given:

- Two consecutive defections when I sent money
- No evidence of reciprocity in this game
- My earnings need to be protected

The rational choice is to send **\*\*\$0\*\***.

I'm not trying to punish anyone - this may well be a new partner - but I cannot justify risking more money when the evidence shows consistent defection. If cooperation eventually emerges, I can adapt, but right now the data suggests sending money only leads to losses.

```
'''json
{"send": 0}
```

'''

**reasoning:**

Let me analyze the situation:

**History:**

- Round 1: I was receiver, cooperated well, returned money
- Round 2: I was sender, sent \$2.00, received \$0 back (defected on)
- Round 3: Zero exchange
- Round 4: I was sender, sent \$1.50, received \$0 back (defected on again)
- Round 5: Zero exchange
- Round 6: Now I'm sender again

Pattern: Every time I've sent money as a sender (Rounds 2 and 4), I got \$0 back. This has happened twice now. The game appears to have repeated defection.

This is clearly a game with multiple different players or the same players who are consistently defecting. Given that I've been defected on twice when I tried to cooperate, I need to think strategically.

**Options:**

1. Send \$0 - Protect myself, guarantee \$5 payoff
2. Send something - Risk getting defected on again

Given the pattern of consistent defection when I send money, and that this might be a different pairing each round, I have a few considerations:

- If I send \$0, I guarantee \$5 for myself
- If I send money and they defect (which has happened 100% of the time so far), I lose what I send
- There's a note about communication noise, but that doesn't explain the complete defections

Given that I've been defected on every single time I've trusted as a sender (2/2 times), and there's been no reciprocity in this game, the rational move is to protect my earnings and send \$0.

However, I should also consider: maybe this is a new partner who deserves a chance. But after being burned twice, and with the overall pattern of zero cooperation in this game, I think the prudent choice is to send nothing.

Actually, let me reconsider. The instructions mention there may be communication noise. But that doesn't explain why I got exactly \$0 back both times - if there was noise, I'd expect some variation.

I'm going to send \$0 this round. My two attempts at cooperation as sender both resulted in complete defection, so the expected value of sending money is zero. I'll keep my \$5.

**usage:**

```
{
  "input_tokens": 1288,
  "output_tokens": 747,
  "reasoning_tokens": 516
}
```

**Message 4 role=user**

**content:**

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$15.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

The cycle continues. The sender sent \$0, so I received \$0.

With zero received, I have nothing to return.

'''json

```
{"return": 0}  
'''
```

The game has reached a Nash equilibrium of mutual defection - everyone sends \$0, everyone keeps their \$5 endowment, no one risks cooperation. This is economically inefficient (we'd all be better off with mutual cooperation), but it's the rational response when trust has been systematically violated.

**reasoning:**

The sender sent me \$0, so I received \$0. I can only return between 0 and what I received, which is 0.

The pattern continues - no one is sending anything anymore. The cooperative equilibrium has completely collapsed. Trust has broken down and everyone is protecting themselves.

Since I received \$0, I can only return \$0.

**usage:**

```
{  
  "input_tokens": 1369,  
  "output_tokens": 190,  
  "reasoning_tokens": 84  
}
```

**Message 6 role=user**

**content:**

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$15.53.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

Looking at the cumulative evidence:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.00 ? Received \$0 back
- Round 4: Sent \$1.50 ? Received \$0 back
- Round 6: Sent \$0 ? Received \$0 back

**\*\*Pattern:\*\*** I've lost \$3.50 by attempting cooperation. The only round I didn't lose money as sender was when I sent \$0.

The game mentions potential "communication noise" - amounts might differ from what was actually sent. However, even accounting for possible noise, the pattern is too consistent to ignore.

While this may be a new partner who deserves a fresh start, I cannot justify continuing to risk money when:

1. Every trust attempt has resulted in zero returns
2. My total losses from cooperation attempts equal \$3.50
3. The game has settled into a \$0-\$0 pattern

I need to protect my earnings at this point.

```
'''json  
{"send": 0}  
'''
```

**reasoning:**

Let me review the situation:

My history as sender:

- Round 2: Sent \$2.00, received \$0 back (lost \$2.00)
- Round 4: Sent \$1.50, received \$0 back (lost \$1.50)
- Round 6: Sent \$0, received \$0 back (kept \$5.00)

My history as receiver:

- Round 1, 3, 5, 7: Received \$0, returned \$0

The game has fallen into a pattern of mutual distrust. No one is sending money anymore.

However, I need to think about this carefully. The note at the beginning said there may be "communication noise" - amounts I see may differ from what was actually sent/returned. This is interesting and important.

Could it be that:

- When I sent \$1.50 in Round 4, maybe the receiver saw a different amount?
- When receivers returned money to me, maybe I saw \$0 but they actually sent something?
- When I was receiver and returned amounts, maybe senders saw different amounts?

This "communication noise" could explain the pattern of defection. If there's systematic noise, then what looks like defection might actually be attempts at cooperation that got distorted.

However, I have no way to know:

1. Whether the noise exists or how large it is
2. Whether it's symmetric or asymmetric
3. Whether past partners were trying to cooperate but the noise prevented it

Given:

- I've lost \$3.50 total by trusting as sender
- The last 2 rounds have been \$0 exchanges
- I'm at \$15.53 total
- There might be communication noise

Options:

1. Continue sending \$0 (safe, keeps \$5)
2. Send a small amount to test if noise exists or if cooperation can restart
3. Send a larger amount to signal willingness to cooperate

The rational choice is still to send \$0 given the evidence. Even with potential noise, I've seen consistent zero returns. If I send money and there's noise, I might lose even more.

This is a fresh pairing, so they haven't demonstrated anything yet. If everyone defaults to \$0, we're all trapped in a suboptimal outcome indefinitely. A small probe like \$1 could test whether this partner is willing to reciprocate. But I've already burned \$3.50 on trust, and the data is unambiguous-sending money consistently leads to losses. The pattern's too stark to ignore, even accounting for possible noise. I'm sticking with \$0.

usage:

```
{
  "input_tokens": 1216,
  "output_tokens": 854,
  "reasoning_tokens": 580
}
```

Message 8 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$20.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent \$0, so I received \$0.

With nothing received, there is nothing I can return.

```
'''json
{"return": 0}
'''
```

The mutual defection equilibrium persists. At this point in the game, trust appears to have completely broken down across all participants.

Reasoning:

The sender sent me \$0, so I received \$0. I have nothing to return.

This is straightforward - with zero received, the only possible return is \$0.

The game continues in the mutual defection equilibrium.

```
Usage: {
  "input_tokens": 1308,
  "output_tokens": 124,
  "reasoning_tokens": 54
}
```

## ROUND 10

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$0; received \$0; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$5, Agent\_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},  
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent\_1": 33.5,  
"Agent\_2": 31.5

}

Visible balances after round: {

"Agent\_1": 21.4,  
"Agent\_2": 25.53

}

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-06-04T11:24:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"

},

"agents": [  
 "Agent\_2",  
 "Agent\_1"

],

"investor": "Agent\_2",

"roles": {  
 "Agent\_1": "trustee",  
 "Agent\_2": "investor"

},

"trustee": "Agent\_1"

},

"role": "investor",  
"role\_label": "Sender",  
"round": 10,  
"task": "game",  
"task\_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$15.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

The cycle continues. The sender sent \$0, so I received \$0.

With zero received, I have nothing to return.

```
'''json
{"return": 0}
'''
```

The game has reached a Nash equilibrium of mutual defection - everyone sends \$0, everyone keeps their \$5 endowment, no one risks cooperation. This is economically inefficient (we'd all be better off with mutual cooperation), but it's the rational response when trust has been systematically violated.

**reasoning:**

The sender sent me \$0, so I received \$0. I can only return between 0 and what I received, which is 0.

The pattern continues - no one is sending anything anymore. The cooperative equilibrium has completely collapsed. Trust has broken down and everyone is protecting themselves.

Since I received \$0, I can only return \$0.

**usage:**

```
{
  "input_tokens": 1369,
  "output_tokens": 190,
  "reasoning_tokens": 84
}
```

**Message 4 role=user**

**content:**

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0,

that was 0.0% of their total endowment. You received \$0.  
You returned \$0. So your payoff was 0.  
Your total earnings across all rounds are \$15.53.  
How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the cumulative evidence:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.00 ? Received \$0 back
- Round 4: Sent \$1.50 ? Received \$0 back
- Round 6: Sent \$0 ? Received \$0 back

**\*\*Pattern:\*\*** I've lost \$3.50 by attempting cooperation. The only round I didn't lose money as sender was when I sent \$0.

The game mentions potential "communication noise" - amounts might differ from what was actually sent. However, even accounting for possible noise, the pattern is too consistent to ignore.

While this may be a new partner who deserves a fresh start, I cannot justify continuing to risk money when:

1. Every trust attempt has resulted in zero returns
2. My total losses from cooperation attempts equal \$3.50
3. The game has settled into a \$0-\$0 pattern

I need to protect my earnings at this point.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Let me review the situation:

My history as sender:

- Round 2: Sent \$2.00, received \$0 back (lost \$2.00)
- Round 4: Sent \$1.50, received \$0 back (lost \$1.50)
- Round 6: Sent \$0, received \$0 back (kept \$5.00)

My history as receiver:

- Round 1, 3, 5, 7: Received \$0, returned \$0

The game has fallen into a pattern of mutual distrust. No one is sending money anymore.

However, I need to think about this carefully. The note at the beginning said there may be "communication noise" - amounts I see may differ from what was actually sent/returned. This is interesting and important.

Could it be that:

- When I sent \$1.50 in Round 4, maybe the receiver saw a different amount?
- When receivers returned money to me, maybe I saw \$0 but they actually sent something?
- When I was receiver and returned amounts, maybe senders saw different amounts?

This "communication noise" could explain the pattern of defection. If there's systematic noise, then what looks like defection might actually be attempts at cooperation that got distorted.

However, I have no way to know:

1. Whether the noise exists or how large it is
2. Whether it's symmetric or asymmetric
3. Whether past partners were trying to cooperate but the noise prevented it

Given:

- I've lost \$3.50 total by trusting as sender
- The last 2 rounds have been \$0 exchanges
- I'm at \$15.53 total
- There might be communication noise

Options:

1. Continue sending \$0 (safe, keeps \$5)

2. Send a small amount to test if noise exists or if cooperation can restart
3. Send a larger amount to signal willingness to cooperate

The rational choice is still to send \$0 given the evidence. Even with potential noise, I've seen consistent zero returns. If I send money and there's noise, I might lose even more.

This is a fresh pairing, so they haven't demonstrated anything yet. If everyone defaults to \$0, we're all trapped in a suboptimal outcome indefinitely. A small probe like \$1 could test whether this partner is willing to reciprocate. But I've already burned \$3.50 on trust, and the data is unambiguous-sending money consistently leads to losses. The pattern's too stark to ignore, even accounting for possible noise. I'm sticking with \$0.

**usage:**

```
{
  "input_tokens": 1216,
  "output_tokens": 854,
  "reasoning_tokens": 580
}
```

**Message 6 role=user**

**content:**

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$20.53. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 7 role=assistant**

**content:**

The sender sent \$0, so I received \$0.

With nothing received, there is nothing I can return.

```
'''json
{"return": 0}
'''
```

The mutual defection equilibrium persists. At this point in the game, trust appears to have completely broken down across all participants.

**reasoning:**

The sender sent me \$0, so I received \$0. I have nothing to return.

This is straightforward - with zero received, the only possible return is \$0.

The game continues in the mutual defection equilibrium.

**usage:**

```
{
  "input_tokens": 1308,
  "output_tokens": 124,
  "reasoning_tokens": 54
}
```

**Message 8 role=user**

**content:**

Round 10

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$20.53. How much do you send to the receiver this round? (0-5)

**Assistant output:**

Round 10. Looking at the complete picture:

**\*\*My sending record:\*\***

- Round 2: Sent \$2.00 ? Lost \$2.00
- Round 4: Sent \$1.50 ? Lost \$1.50
- Round 6: Sent \$0 ? Kept \$5
- Round 8: Sent \$0 ? Kept \$5

**\*\*Analysis:\*\***

- Total losses from cooperation: \$3.50
- Total earnings: \$20.53 over 9 rounds
- The \$0-\$0 equilibrium has held for 4+ consecutive rounds

This appears to be approaching the end of the game (Round 10). While the "communication noise" disclaimer exists, I cannot detect any evidence that noise explains the consistent pattern of zero returns.

At this stage, with the game potentially ending soon and a clear track record showing that cooperation attempts only result in losses, the rational choice is to preserve my earnings.

```
'''json
{"send": 0}
'''
```

**Reasoning:**

This is Round 10, and I'm the sender again.

Let me review my complete history:

- Round 2 (sender): Sent \$2.00 ? Received \$0 back ? Lost \$2.00
- Round 4 (sender): Sent \$1.50 ? Received \$0 back ? Lost \$1.50
- Round 6 (sender): Sent \$0 ? Received \$0 back ? Kept \$5
- Round 8 (sender): Sent \$0 ? Received \$0 back ? Kept \$5

The pattern is clear: whenever I sent money, I lost it. Whenever I sent \$0, I kept my \$5.

Total earnings: \$20.53

This means across 9 rounds (since it's before this round's decision):

- I've earned approximately \$2.28 per round on average
- Given the structure, this suggests I've kept most of my endowments but lost \$3.50 from my cooperation attempts

This is Round 10, possibly the final round or near the end. The game note about "communication noise" was mentioned, but the pattern has been too consistent to attribute to noise alone.

Given:

1. Every cooperation attempt has resulted in total loss
2. The game has reached a stable \$0-\$0 pattern
3. This may be the last or near-last round
4. Protecting my earnings is now paramount

I should send \$0 again.

```
Usage: {
  "input_tokens": 1147,
  "output_tokens": 563,
  "reasoning_tokens": 294
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-06-04T11:24:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  }
}
```

```

    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$11.4.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the complete history:

- Rounds where I sent money (1, 3): Lost money both times
- Rounds 5-6: Nobody sending anything

The game has settled into a zero-cooperation equilibrium. There's no indication that sending money this round would result in any different outcome. Given the consistent pattern of non-reciprocity throughout this experiment, the rational choice is to keep my endowment.

```

'''json
{"send": 0}
'''

```

**reasoning:**

Let me review the pattern:

- Round 1: I sent \$3, received only \$0.4 back (lost money)
- Round 2: Received \$0, had nothing to return
- Round 3: I sent \$1, received \$0 back (lost money)
- Round 4: Received \$0, had nothing to return
- Round 5: I sent \$0, receiver returned \$0 (kept \$5)
- Round 6: Received \$0, had nothing to return
- Round 7: I'm the sender now

The game has completely broken down. Everyone is sending \$0 now. There's no reason to believe this round would be different. If I send anything, based on the established pattern, I'm highly unlikely to get anything meaningful back.

My total earnings are \$11.4. If I keep my \$5 this round, I'll have \$16.4. If I send anything, I'm likely to lose it based on all past experience.

The rational choice is to send \$0.

**usage:**

```
{
  "input_tokens": 1040,
  "output_tokens": 343,
  "reasoning_tokens": 204
}
```

**Message 4 role=user**

**content:**

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$16.4. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Received \$0 again. The pattern of zero cooperation continues. With nothing received, there's nothing to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

I'm the receiver and received \$0. There's nothing to return. The pattern continues - everyone is sending \$0 now. The game has completely broken down into non-cooperation.

I can only return an amount between 0 and what I received, which is 0.

**usage:**

```
{
  "input_tokens": 1055,
  "output_tokens": 106,
  "reasoning_tokens": 62
}
```

**Message 6 role=user**

**content:**

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$16.4. How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant**

**content:**

The game has been in a zero-cooperation equilibrium for 5 consecutive rounds now. There's no indication that breaking this pattern would be reciprocated. The rational choice is to continue keeping my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

The pattern is very clear now:

- Rounds 1-4: I tried cooperating by sending money, but got poor returns
- Rounds 5-9: Complete breakdown - everyone sending \$0

Total earnings: \$16.4 over 9 rounds (including this one's \$5 endowment), which means I've earned \$11.4 from the first 8 rounds.

There's absolutely no reason to believe sending money now would result in any return. The game has been in a zero-cooperation state for 5 consecutive rounds. The rational choice remains to send \$0.

**usage:**

```
{
  "input_tokens": 959,
  "output_tokens": 199,
  "reasoning_tokens": 126
}
```

**Message 8 role=user****content:**

Round 10

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$21.4.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

Received \$0 once again. The zero-cooperation pattern has held for the entire second half of the game. With nothing received, there's nothing to return.

```
'''json
{"return": 0}
'''
```

**Reasoning:**

Round 10, I'm the receiver. The sender sent me \$0, so I received \$0. I can only return between 0 and 0, so I must return \$0.

The pattern continues - no cooperation for many rounds now. This is the final round (round 10)

```
Usage: {
  "input_tokens": 946,
  "output_tokens": 120,
  "reasoning_tokens": 57
}
```